

Distributed Storage Systems with Quantum Entanglement Assistance

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Abstract

We study the potential advantages enabled by quantum resources in distributed storage systems. Consider an (n, k, d) distributed storage system in which a file is stored across n nodes such that any k nodes suffice to reconstruct the file. When a node fails, any d helper nodes transmit information to a newcomer to rebuild the system. In contrast to the classical repair, where helper nodes transmit classical bits, we allow them to send classical information over *quantum channels* to the newcomer. The newcomer then generates its storage by performing appropriate measurements on the received quantum states. Within this framework, we fully characterize the fundamental tradeoff between classical storage and quantum repair bandwidth (total communication cost). We present converse and achievability proofs to show the optimality of the tradeoff. The converse proof relies on dimension bound and data processing inequality from quantum information theory, while the achievability proof develops a structured invariant-based framework that guarantees preservation of the data retrieval property under arbitrary repair sequences. Our results demonstrate that compared to classical systems, the optimal storage-bandwidth tradeoff can be *significantly improved* with the enhancement of quantum entanglement shared only among the surviving nodes, particularly at the minimum-storage regenerating point. Remarkably, we show that when $d \geq 2k - 2$, there exists an operating point at which *both storage and repair bandwidth are simultaneously minimized*. This phenomenon breaks the tradeoff in the classical setting and reveals a fundamentally new regime enabled by quantum communication.

1 Introduction

Distributed storage systems are a fundamental infrastructure for modern data centers and cloud storage platforms [1–3]. Since individual storage nodes may be unreliable or subject

to erasures, redundancy must be introduced to ensure data reliability. Specifically, a file of size B is encoded and stored across n nodes such that the original file can be reconstructed by accessing the contents of any k nodes (referred to as the *data retrieval* property) [4, 5]. A widely adopted solution is to employ an (n, k) maximum distance separable (MDS) code, which achieves the minimum possible per-node storage of B/k .

While MDS codes are optimal from a storage-efficiency perspective, they may incur a substantial communication cost during node repair. When a storage node fails, the system initiates a repair process to preserve the data retrieval property. Specifically, a newcomer node connects to any d surviving helper nodes, each of which transmits β dits of information. Based on the received data, the newcomer generates its stored content so that the system continues to satisfy the data retrieval requirement.¹ A fundamental tradeoff arises between storage and repair bandwidth. For instance, if an MDS code with per-node storage B/k is used and $d = k$, repairing a failed node requires downloading the entire file, resulting in repair bandwidth B . However, by allowing larger per-node storage, the repair bandwidth can be reduced. In fact, increasing the per-node storage to $2B/(k + 1)$ reduces the repair bandwidth to $2B/(k + 1)$. This tradeoff was fully characterized by Dimakis *et al.* [4], which showed that per-node storage and repair bandwidth cannot, in general, be simultaneously minimized. Thus, reducing storage necessarily increases repair bandwidth, and vice versa.

Over the past decades, quantum technology has been shown to enhance the efficiency of classical communication significantly [6–8]. A typical example is superdense coding, in which a single qubit, when assisted by pre-shared entanglement between the transmitter and the receiver, can convey up to two classical bits of information. More recently, quantum resources have also demonstrated significant advantages in distributed information processing tasks such as private information retrieval (PIR) [9–13] and distributed computation [14–17]. In these settings, entanglement is shared only among static entities such as servers (transmitters), which is more realistic in practice than entanglement with dynamic users. In PIR settings, a user wishes to download a classical message stored at the servers in such a way that the servers do not learn which particular message the user aims to download. Classical schemes necessarily introduce structured interference to mask the user’s request, increasing the total download cost. In contrast, quantum resources allow the user to eliminate part of the interference subspace via appropriate quantum measurements, thereby improving the achievable rate. It was shown in various PIR variants that quantum schemes achieve up to twice of the classical rate. Similarly, in distributed computation problems, a user seeks to compute a function (e.g., summation) of classical data stored across multiple servers. Shared entanglement among the servers can reduce the required communication cost. In particular, for certain linear computations, the communication cost can be reduced by a

¹The repair problem has two variants: *exact repair* and *functional repair*. In exact repair, the newcomer stores exactly the same content as the failed node, whereas in functional repair, the stored content may differ as long as the data retrieval property is preserved. In this paper, we focus on the more general functional repair setting.

factor of two compared to classical schemes [14, 15]. We note here that distributed storage problem is fundamentally different from distributed computation problems. In particular, in each repair process, the newcomer is not required to compute a specific function of the storage of the helper nodes. Instead, it only needs to generate storage such that the global data retrieval property is preserved. This additional structural flexibility raises the possibility that quantum communication can alter the fundamental performance limits of distributed storage systems.

Motivated by this, we seek to characterize the new storage-bandwidth tradeoff when helper nodes are allowed to transmit quantum states (qudits), and to determine the maximum performance gains achievable via entanglement assistance among the surviving nodes (transmitters). We consider an entanglement-assisted repair model in which the helper nodes share a pre-established entangled quantum state and locally encode their stored classical information into their states, which are then transmitted to the newcomer. Upon receiving these quantum states, the newcomer performs appropriate measurements to generate its stored content, while ensuring that the resulting storage system continues to satisfy the data retrieval property. In this framework, we fully characterize the fundamental tradeoff between classical storage and quantum repair bandwidth. We provide converse and achievability results that prove the optimality of the tradeoff. The converse proof relies on quantum information-theoretic inequalities. The achievability proof develops a structured invariant-based framework, ensuring that the data retrieval property is preserved under arbitrary sequences of repair rounds. Furthermore, we analyze the performance gains enabled by quantum resources at both the minimum-storage regenerating (MSR) point and the minimum-bandwidth regenerating (MBR) point, and provide a pictorial explanation to explicitly show the quantum advantage regions. Our results show that when $d \leq 2k - 2$, entanglement-assisted repair can reduce the required repair bandwidth by a factor of two at the MSR point. Further, when $d \geq 2k - 2$, we identify a unique operating point at which both the repair bandwidth and per-node storage are *simultaneously minimized*, which does not occur in classical systems.

Notation: For an integer M , $[M]$ denotes the set $\{1, \dots, M\}$. For integers a and b , $[a : b]$ denotes $\{a, \dots, b\}$. $A_{[M]}$ is the compact notation of $\{A_1, A_2, \dots, A_M\}$. $\otimes$ denotes the tensor product. For a q -dimensional quantum system, the computational basis is denoted as $\{|j\rangle\}_{j \in \mathbb{F}_q}$, where $q \triangleq p^r$ and p is a prime number. For a vector $\mathbf{u} = [u_1, \dots, u_n]^T$, we use $\text{diag}(\mathbf{u})$ to denote the $n \times n$ diagonal matrix whose i th diagonal entry is u_i , $i \in [n]$.

2 Preliminaries

2.1 Classical Error Correcting Codes

Definition 1 (Generalized Reed-Solomon (GRS) Code) Let $\mathbb{F}_q$ be a finite field, and let $\boldsymbol{\alpha} \triangleq [\alpha_1, \dots, \alpha_n] \in \mathbb{F}_q^n$ consist of n distinct elements and $\mathbf{u} \triangleq [u_1, \dots, u_n] \in \mathbb{F}_q^n$ consist of n nonzero elements. Assume that $q \geq n$. The Generalized Reed-Solomon (GRS) code $\mathcal{C}$ is the length- n , dimension- k linear code over $\mathbb{F}_q$ defined as

$$\mathcal{C} \triangleq \left\{ (u_1 f(\alpha_1), \dots, u_n f(\alpha_n)) : f(x) \in \mathbb{F}_q[x], \deg f < k \right\}. \quad (1)$$

A generator matrix of $\text{GRS}_{k,n}^q(\boldsymbol{\alpha}, \mathbf{u})$ is given by

$$\text{GRS}_{k,n}^q(\boldsymbol{\alpha}, \mathbf{u}) \triangleq \begin{bmatrix} u_1 & u_1 \alpha_1 & u_1 \alpha_1^2 & \dots & u_1 \alpha_1^{k-1} \\ u_2 & u_2 \alpha_2 & u_2 \alpha_2^2 & \dots & u_2 \alpha_2^{k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ u_n & u_n \alpha_n & u_n \alpha_n^2 & \dots & u_n \alpha_n^{k-1} \end{bmatrix} \in \mathbb{F}_q^{n \times k}. \quad (2)$$

It is known that GRS codes are maximum-distance separable (MDS), i.e., any k rows of $\text{GRS}_{k,n}^q(\boldsymbol{\alpha}, \mathbf{u})$ are linearly independent, and the code distance satisfies $d = n - k + 1$.

Moreover, the dual of a GRS code is also a GRS code. Specifically, there exists a vector $\mathbf{v} = [v_1, \dots, v_n] \in \mathbb{F}_q^n$ with $v_i \neq 0$ such that [11]

$$\text{GRS}_{k,n}^q(\boldsymbol{\alpha}, \mathbf{u})^\perp = \text{GRS}_{n-k,n}^q(\boldsymbol{\alpha}, \mathbf{v}). \quad (3)$$

Equivalently, the corresponding generator matrices satisfy

$$(\text{GRS}_{k,n}^q(\boldsymbol{\alpha}, \mathbf{u}))^T \cdot \text{GRS}_{k,n-k}^q(\boldsymbol{\alpha}, \mathbf{v}) = \mathbf{0}_{k \times (n-k)}. \quad (4)$$

Finally, the generator matrix admits the decomposition

$$\text{GRS}_{k,n}^q(\boldsymbol{\alpha}, \mathbf{u}) = \text{diag}(\mathbf{u}) \underbrace{\begin{bmatrix} 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{k-1} \\ 1 & \alpha_2 & \alpha_2^2 & \dots & \alpha_2^{k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \dots & \alpha_n^{k-1} \end{bmatrix}}_{\triangleq \mathbf{A}}, \quad (5)$$

where $\mathbf{A}$ is a Vandermonde matrix.

2.2 Quantum Information

Definition 2 (Quantum channel and Kraus representation [18]) *A quantum channel with input quantum system A and output quantum system B is a completely positive and trace-preserving (CPTP) linear mapping*

$$\Lambda : \mathcal{D}(\mathcal{H}_A) \rightarrow \mathcal{D}(\mathcal{H}_B), \quad (6)$$

where $\mathcal{D}(\mathcal{H})$ denotes the set of density operators on the Hilbert space $\mathcal{H}$. Moreover, Λ can be represented by a Kraus representation

$$\Lambda(\rho) = \sum_k K_k \rho K_k^\dagger, \quad (7)$$

where $\{K_k\}$ are Kraus operators satisfying

$$\sum_k K_k^\dagger K_k = I, \quad (8)$$

where I is the identity matrix.

Definition 3 (Generalized Pauli operators) *For any $a, b \in \mathbb{F}_q$, define two single-qudit Pauli operators as*

$$\mathbf{X}(a) \triangleq \sum_{j \in \mathbb{F}_q} |j+a\rangle \langle j|, \quad \mathbf{Z}(b) \triangleq \sum_{j \in \mathbb{F}_q} \omega^{\text{tr}(bj)} |j\rangle \langle j|, \quad (9)$$

where $\omega \triangleq \exp(2\pi\sqrt{-1}/p)$ and $\text{tr} : \mathbb{F}_q \rightarrow \mathbb{F}_p$ denotes the field trace. For vectors $\mathbf{x} \triangleq [x_1, \dots, x_n]^T$, $\mathbf{z} \triangleq [z_1, \dots, z_n]^T \in \mathbb{F}_q^{n \times 1}$, the corresponding n -qudit Pauli operator is defined as the tensor product

$$\mathbf{X}(\mathbf{x})\mathbf{Z}(\mathbf{z}) \triangleq \bigotimes_{i \in [n]} \mathbf{X}(x_i)\mathbf{Z}(z_i). \quad (10)$$

Definition 4 (CSS codes) *A CSS code $\mathcal{C} = \text{CSS}(\mathcal{C}_X, \mathcal{C}_Z)$ is a stabilizer code² defined by two classical linear codes $\mathcal{C}_X, \mathcal{C}_Z \subseteq \mathbb{F}_q^n$ satisfying $\mathcal{C}_X^\perp \subseteq \mathcal{C}_Z$. The corresponding stabilizer group is specified by*

$$\mathcal{S} \triangleq \{\mathbf{X}(\mathbf{a})\mathbf{Z}(\mathbf{b}) \mid \mathbf{a} \in \mathcal{C}_X^\perp, \mathbf{b} \in \mathcal{C}_Z^\perp\}. \quad (11)$$

Definition 5 (Syndrome measurement for CSS codes [10]) *A general quantum measurement is described by a positive operator-valued measure (POVM), which is a set $\{Q_i\}_{i \in M}$*

²A stabilizer code is a code that can be defined by its stabilizer group only and whose codewords are the +1 eigenvectors of the stabilizer group [18].

of positive semidefinite operators satisfying $\sum_{i \in M} Q_i = I$.

For a CSS code $\text{CSS}(\mathcal{C}_X, \mathcal{C}_Z)$, the syndrome for a Pauli error $\mathbf{X}(\mathbf{x})\mathbf{Z}(\mathbf{z})$ is obtained by measuring the stabilizer generators. Let $\mathbf{H}_X$ and $\mathbf{H}_Z$ be the parity-check matrices of $\mathcal{C}_X$ and $\mathcal{C}_Z$, respectively. The syndrome measurement outcome is expressed by [19]

$$\mathbf{s}_X = \mathbf{H}_Z^T \mathbf{x}, \quad \mathbf{s}_Z = \mathbf{H}_X^T \mathbf{z}. \quad (12)$$

Remark 1 Syndrome measurement provides the mechanism for connecting classical information with quantum states in error correction. By encoding classical data into the Pauli parameters $(\mathbf{x}, \mathbf{z})$ and applying the corresponding Pauli operator $\mathbf{X}(\mathbf{x})\mathbf{Z}(\mathbf{z})$ to a quantum state, the receiver can extract the syndromes $\mathbf{s}_X, \mathbf{s}_Z$. One can therefore design the classical codes $\mathcal{C}_X, \mathcal{C}_Z$ so that these syndromes are specific linear functions of $(\mathbf{x}, \mathbf{z})$. This enables the construction of our achievability scheme.

3 Problem Statement & Classical Results

3.1 System Model

We consider an (n, k, d) distributed storage system storing a (classical) file consisting of B dits across n storage nodes, where each node stores α dits. The system is required to satisfy the following properties:

- **(Data retrieval)** The entire file can be reconstructed by a data collector (DC) from the contents of any k nodes.
- **(Single-node repair)** Upon the failure of a storage node, a newcomer node connects to any d surviving nodes among the remaining $n - 1$ nodes, where $k \leq d \leq n - 1$ [20].³ After the repair process, the resulting system must continue to satisfy the data retrieval property.

We consider the repair process under classical and quantum communication models specified as follows.

3.1.1 Classical Repair

In the classical setting, each of the d surviving nodes transmits β_c dits to the newcomer, resulting in a total repair communication cost (repair bandwidth) of $d\beta_c$. A fundamental result for this $(n, k, d, \alpha, \beta_c, B)$ system is that there exists a tradeoff between the per-node storage α and the repair bandwidth $d\beta_c$: increasing storage can reduce the required repair

³We assume that k and d are chosen minimally such that the data retrieval property and single-node repair property hold. In particular, if $d < k$, then the file could be reconstructed from fewer than k nodes, contradicting the minimality of k [20].

bandwidth, and vice versa. This storage-bandwidth tradeoff was fully characterized in [4], and is reviewed in Section 3.2.

3.1.2 Quantum Repair

In this paper, we explore performance improvement by allowing *quantum communication* during the repair process. Specifically, the d helper nodes are assumed to share a prior entangled state. Each helper node transmits β_q qudits to the newcomer. By performing appropriate quantum measurements on the received quantum systems, the newcomer reconstructs α classical dits to be stored locally, such that the repaired system continues to satisfy the *data retrieval* property.

The quantum communication-assisted repair steps are specified as follows.

- a) **(Entanglement distribution)** Suppose the newcomer node communicates with d nodes among the $n - 1$ surviving nodes, indexed by $s'_1, \dots, s'_d$. These d nodes share a globally entangled quantum state represented by the density matrix ρ^{ini} in the quantum system $\mathcal{Q} = \mathcal{Q}_1 \cdots \mathcal{Q}_d$. This state is pre-established before the communication begins. The dimension of each quantum subsystem $\mathcal{Q}_s$ is given by $\delta_s \triangleq |\mathcal{Q}_s|$, and each subsystem consists of β_q qudits, i.e., $\log_q \delta_s = \beta_q$.
- b) **(Information encoding)** Each helper node $s \in \{s'_1, \dots, s'_d\}$ applies a local encoding operation to its quantum state on its subsystem $\mathcal{Q}_s$ in order to embed classical information determined by its stored data. Specifically, node s applies a CPTP map Λ_s to its quantum state based on its local storage content. After all encoding operations are applied, the joint quantum state is given by the density matrix

$$\rho^{\text{enc}} = \bigotimes_{s \in \{s'_1, \dots, s'_d\}} \Lambda_s(\rho^{\text{ini}}). \quad (13)$$

The d encoded quantum subsystems are then transmitted to the newcomer through separate *noiseless* quantum channels.

- c) **(Storage generation)** Upon receiving the encoded quantum state, ρ^{enc} , the newcomer performs a suitable POVM. The resulting measurement outcomes allow the newcomer to generate α dits as its storage, such that the repaired system continues to satisfy the *data retrieval* property and the *single-node repair* property.

Our objective is to characterize the new tradeoff between storage and repair bandwidth for this $(n, k, d, \alpha, \beta_q, B)$ system. We compare the tradeoffs in the classical and quantum settings in order to show the potential quantum advantage in distributed storage systems. To this end, we next review the result in classical distributed storage systems.

3.2 Classical Results

The fundamental storage-repair bandwidth (in dits) tradeoff was established in [4]. Specifically, by applying the cut-set bound from network coding, any classical distributed storage system with parameters $(n, k, d, \alpha, \beta_c, B)$ must satisfy

$$[\text{Classical}] : \sum_{i=0}^{k-1} \min\{(d-i)\beta_c, \alpha\} \geq B. \quad (14)$$

This inequality characterizes all achievable tradeoff between per-node storage α and repair bandwidth $d\beta_c$.

Two extremal points on the tradeoff curve are of particular interest: the *minimum-storage regenerating* (MSR) point and the *minimum-bandwidth regenerating* (MBR) point [4, 20], which are specified as follows.

- The MSR point is obtained by first minimizing the per-node storage α and then minimizing the per-node bandwidth β_c . The corresponding operating point is

$$(\alpha^{\text{MSR}}, d\beta_c^{\text{MSR}}) = \left(\frac{B}{k}, \frac{B}{k} \cdot \frac{d}{d-k+1} \right). \quad (15)$$

- The MBR point is obtained by first minimizing β_c and then minimizing α . The resulting operating point is

$$(\alpha^{\text{MBR}}, d\beta_c^{\text{MBR}}) = \left(\frac{2Bd}{2kd - k^2 + k}, \frac{2Bd}{2kd - k^2 + k} \right). \quad (16)$$

Remark 2 Note that $\alpha^{\text{MBR}} \geq \alpha^{\text{MSR}}$ and the equality holds if and only if the file is replicated among all nodes ($k = 1$). Hence, for any non-replicated storage system, i.e., $k \geq 2$, the MSR point and MBR point do not coincide in the classical setting. In the following, we show that with the assistance of quantum communication and entanglement, it is possible to simultaneously minimize both the storage and repair bandwidth (in qudits).

4 Main Results

Theorem 1 Consider a distributed storage system with parameters $(n, k, d, \alpha, \beta_q, B)$. The optimal tradeoff between storage and repair bandwidth (measured in qudits) is characterized by

$$[\text{Quantum}] : \sum_{i=0}^{k-1} \min\{2(d-i)\beta_q, d\beta_q, \alpha\} \geq B. \quad (17)$$

The converse and achievability proofs of Theorem 1 are presented in Section 6 and Section 7, respectively. We now provide more intuition on Theorem 1. The bound in Theorem 1 can be interpreted via a cut-set argument on a quantum-communication-assisted information flow graph. To be specific, the information flow graph consists of a single data source $\mathbf{S}$, storage nodes $\mathbf{x}_{\text{in}}^i, \mathbf{x}_{\text{out}}^i$ for each $i \in [n]$, and a data collector $\mathbf{DC}$. The source node $\mathbf{S}$ represents the original file. Each storage node i is modeled by a pair of vertices $\mathbf{x}_{\text{in}}^i$ and $\mathbf{x}_{\text{out}}^i$ connected by a directed edge with capacity equal to the storage size at node i . A data collector $\mathbf{DC}$ reconstructs the file by connecting to any k storage nodes through edges with infinite capacities. Here, a cut is defined as a set of edges whose removal separates $\mathbf{S}$ from $\mathbf{DC}$, and the minimum cut is the cut with the smallest total capacity among all such separating cuts.

Now, consider a distributed storage system with parameters $(n, k, d) = (4, 2, 3)$, storage $\alpha = 3$, and per-helper bandwidth $\beta_c = \beta_q \triangleq \beta = 1$. The classical cutset bound (14) gives

$$\sum_{i=0}^1 \min\{(3-i), 3\} = 3 + 2 = 5, \quad (18)$$

which corresponds to the cut illustrated in Fig. 1. Hence, the system can store a file of size at most $B = 5$ dits. In contrast, with quantum communication and pre-shared entanglement among the helper nodes, the corresponding bound in (17) becomes

$$\sum_{i=0}^1 \min\{2(3-i), 3, 3\} = 3 + 3 = 6, \quad (19)$$

which corresponds to the same cut in Fig. 1 with increased effective capacity.

This gain arises because, while the classical cut consists of one storage edge of capacity $\alpha = 3$ and two repair edges each of capacity $\beta = 1$, the quantum communication links can jointly convey additional information. In particular, due to the superdense coding gain, two quantum links carrying a total of 2β qudits can transmit up to 4β dits of classical information [7], while the Holevo bound [21] requires that the total amount of information is restricted by the total number of transmitted qudits from the 3 helper nodes, i.e., 3β . Hence, the two quantum links can send $\min\{4\beta, 3\beta\} = 3$ dits. As a result, the entanglement-assisted system provides a strictly larger cut capacity than its classical counterpart.

From Theorem 1, the quantum MSR (QMSR) point and the quantum MBR (QMBR) point can be derived explicitly in the following corollaries.

Corollary 1 *The QMSR point is given by the pair*

$$(\alpha^{\text{QMSR}}, d\beta_q^{\text{QMSR}}) = \left(\frac{B}{k}, \frac{B}{k} \max \left\{ 1, \frac{d}{2(d-k+1)} \right\} \right) \quad (20)$$

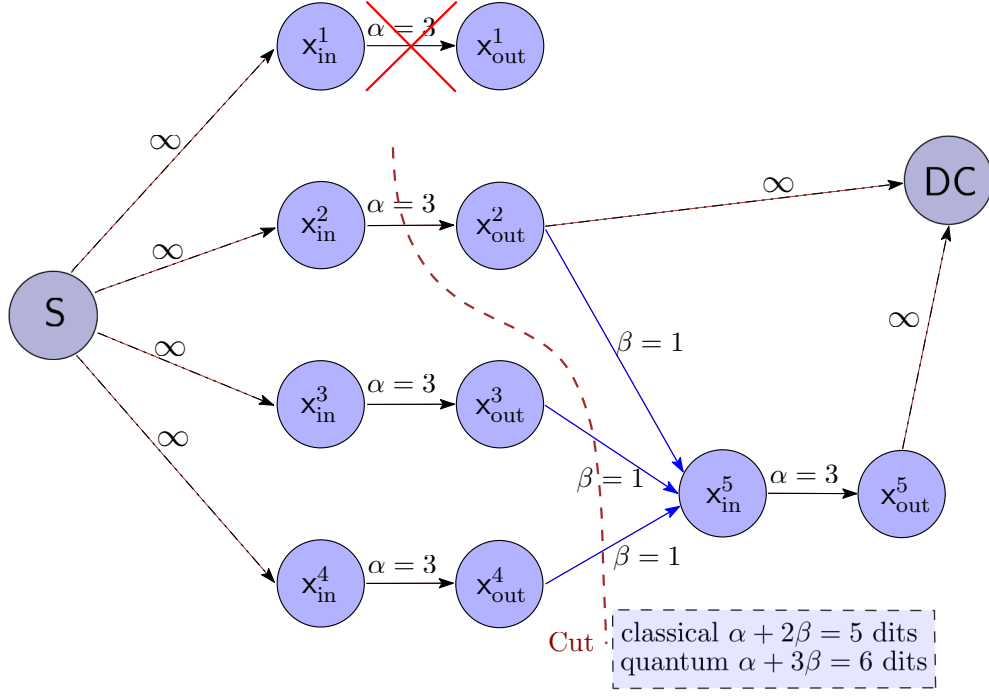


Figure 1: The classical and quantum cut for distributed storage system with $(n, k, d) = (4, 2, 3)$, $\alpha = 3$, and $\beta_c = \beta_q = \beta = 1$.

$$= \begin{cases} \left(\frac{B}{k}, \frac{B}{k} \right), & \text{if } d \geq 2k - 2, \\ \left(\frac{B}{k}, \frac{Bd}{2k(d - k + 1)} \right), & \text{if } k \leq d < 2k - 2. \end{cases} \quad (21)$$

Proof: We first minimize the storage parameter α subject to the cut-set bound in (17). The minimum storage is achieved when the term α is the active constraint in every summand, i.e., $\alpha \leq \min\{2(d - i)\beta_q, d\beta_q\}, \forall i = 0, \dots, k - 1$, i.e.,

$$\alpha \leq \beta_q \cdot \min\{d, 2(d - k + 1)\}. \quad (22)$$

Under this condition, (17) reduces to $k\alpha \geq B$, and hence the minimum storage is $\alpha^{\text{QMSR}} = \frac{B}{k}$. Next, we minimize β_q subject to (22). Substituting α_{QMSR} into (22) gives

$$\beta_q^{\text{QMSR}} = \frac{B}{k} \cdot \frac{1}{\min\{d, 2(d - k + 1)\}} \quad (23)$$

$$= \frac{B}{k} \cdot \max\left\{ \frac{1}{d}, \frac{1}{2(d - k + 1)} \right\}. \quad (24)$$

Multiplying both sides by d yields the stated expression for $d\beta_{\text{QMSR}}$, completing the proof. ■

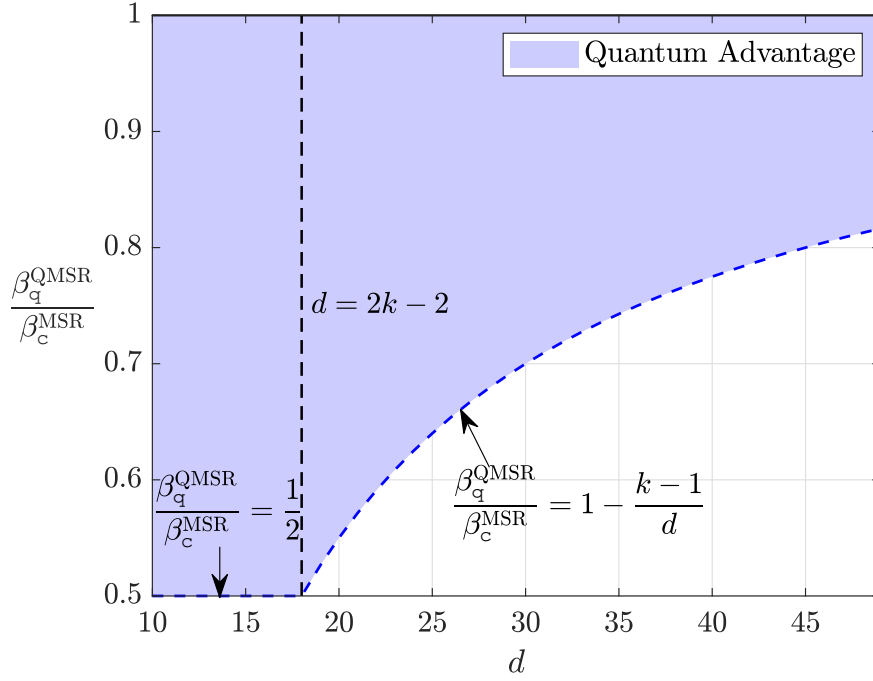


Figure 2: The per-node bandwidth ratio $\frac{\beta_q^{\text{QMSR}}}{\beta_c^{\text{MSR}}}$ versus the number of helper nodes d , when $n = 50, k = 10$.

Remark 3 Compared with the classical MSR point, the minimum storage remains unchanged, i.e., $\alpha^{\text{MSR}} = \alpha^{\text{QMSR}} = \frac{B}{k}$. However, the required bandwidth with quantum repair is strictly smaller. In particular, the ratio between the quantum and classical per-helper bandwidths is given by

$$\frac{\beta_q^{\text{QMSR}}}{\beta_c^{\text{MSR}}} = \max \left\{ 1 - \frac{k-1}{d}, \frac{1}{2} \right\} \quad (25)$$

$$= \begin{cases} 1 - \frac{k-1}{d}, & \text{if } d \geq 2k-2, \\ \frac{1}{2}, & \text{if } k \leq d < 2k-2. \end{cases} \quad (26)$$

This ratio is illustrated in Fig. 2. Observe that the bandwidth advantage enabled by quantum resources is maximized in the regime of small repair degree, i.e., $d \leq 2k-2$, where the quantum bandwidth is exactly one half of its classical counterpart. As d increases beyond this threshold, the relative gain gradually diminishes and vanishes asymptotically as $d \rightarrow \infty$. This is because, for large repair degrees, d , the classical repair process already approaches its optimal efficiency, i.e., $\frac{d\beta_c^{\text{MSR}}}{\alpha^{\text{MSR}}} \rightarrow 1$, leaving limited room for further improvement via quantum communication.

Corollary 2 *The QMBR point is given by the pair*

$$(\alpha^{\text{QMBR}}, d\beta_{\mathbf{q}}^{\text{QMBR}}) = \begin{cases} \left(\frac{B}{k}, \frac{B}{k}\right), & \text{if } d \geq 2k - 2, \\ (\bar{\alpha}, \bar{\alpha}), & \text{if } k \leq d < 2k - 2, \end{cases} \quad (27)$$

where

$$\bar{\alpha} \triangleq \frac{dB}{d(\lfloor d/2 \rfloor + 1) + (2d - k - \lfloor d/2 \rfloor)(k - \lfloor d/2 \rfloor - 1)}. \quad (28)$$

Proof: We first minimize the repair bandwidth parameter $\beta_{\mathbf{q}}$, and then minimize the storage parameter α . From the bound in (17), the minimum $\beta_{\mathbf{q}}$ is attained when the bandwidth-related terms are active in each summand, namely,

$$\min\{d\beta_{\mathbf{q}}, 2(d-i)\beta_{\mathbf{q}}\} \leq \alpha, \quad i = 0, 1, \dots, k-1. \quad (29)$$

Under this condition, the bound (17) reduces to

$$B = \beta_{\mathbf{q}} \sum_{i=0}^{k-1} \min\{d, 2(d-i)\}. \quad (30)$$

We analyze this expression by considering the following two cases.

1. If $d \leq 2(d-i), \forall i = 0, \dots, k-1$, i.e., $d \geq 2k-2$, then

$$B = kd\beta_{\mathbf{q}}, \quad (31)$$

which implies $\alpha_{\text{QMBR}} = d\beta_{\mathbf{q}}^{\text{QMBR}} = \frac{B}{k}$.

2. If $d > 2(d-k+1)$, i.e., $d < 2k-2$, then the minimum in $\min\{d, 2(d-i)\}$ switches at $i = \lfloor d/2 \rfloor$. Specifically,

$$\min\{d, 2(d-i)\} = \begin{cases} d, & i = 0, 1, \dots, \lfloor d/2 \rfloor, \\ 2(d-i), & i = \lfloor d/2 \rfloor + 1, \dots, k-1. \end{cases} \quad (32)$$

Hence, we have

$$B = (\lfloor d/2 \rfloor + 1)d\beta_{\mathbf{q}} + 2 \sum_{i=\lfloor d/2 \rfloor + 1}^{k-1} (d-i)\beta_{\mathbf{q}} \quad (33)$$

$$= (\lfloor d/2 \rfloor + 1)d\beta_{\mathbf{q}} + (2d - k - \lfloor d/2 \rfloor)(k - \lfloor d/2 \rfloor - 1)\beta_{\mathbf{q}}. \quad (34)$$

Solving for $\beta_{\mathbf{q}}$ yields $d\beta_{\mathbf{q}}^{\text{QMBR}}$. This gives the expression of the QMBR point for $k \leq$

$$d \leq 2k - 2.$$

This completes the proof. ■

Remark 4 We now compare the QMBR point with the classical MBR point. Since both points satisfy $\alpha = d\beta$, it suffices to compare the ratio of the per-helper repair bandwidths $\frac{\beta_q^{\text{QMBR}}}{\beta_c^{\text{MBR}}}$.

- When $d \geq 2k - 2$, the ratio between the quantum and classical per-helper repair bandwidths is given by

$$\frac{\beta_q^{\text{QMBR}}}{\beta_c^{\text{MBR}}} = 1 - \frac{k-1}{2d}. \quad (35)$$

Hence,

$$\frac{3}{4} \leq \frac{\beta_q^{\text{QMBR}}}{\beta_c^{\text{MBR}}} \leq 1. \quad (36)$$

Moreover, for a fixed k , this ratio is a monotonically increasing function of d , rising from $3/4$ and approaching 1 as d grows. This indicates that the relative bandwidth reduction offered by quantum communication becomes less significant when the repair degree is much larger than k .

- When $k \leq d \leq 2k - 2$, the ratio between the quantum and classical per-helper bandwidth is

$$r(d, k) \triangleq \frac{\beta_q^{\text{QMBR}}}{\beta_c^{\text{MBR}}} = \frac{k(2d - k + 1)}{2d(\lfloor d/2 \rfloor + 1) + 2(2d - k - \lfloor d/2 \rfloor)(k - \lfloor d/2 \rfloor - 1)}, \quad (37)$$

which depends on whether d is odd or even. Nevertheless, several qualitative properties can be observed as follows.

1. For a fixed k , the ratio $r(d, k)$ increases monotonically with d over the interval $k \leq d \leq 2k - 2$, and it reaches $3/4$ when $d = 2k - 2$.
2. Consider three consecutive values of d , namely $d = 2m$, $2m + 1$, and $2m + 2$. The increment of the ratio from the even value $2m$ to the subsequent odd value $2m + 1$ is no smaller than that from $2m + 1$ to $2m + 2$, i.e.,

$$r(2m + 1, k) - r(2m, k) \geq r(2m + 2, k) - r(2m + 1, k). \quad (38)$$

The proof is presented in Appendix A.

Overall, the per-node bandwidth ratio $\frac{\beta_q^{\text{QMBR}}}{\beta_c^{\text{MBR}}}$ is a monotonically increasing function of the number of helper nodes d over the entire range $k \leq d \leq n - 1$. In particular, when $k \leq d \leq$

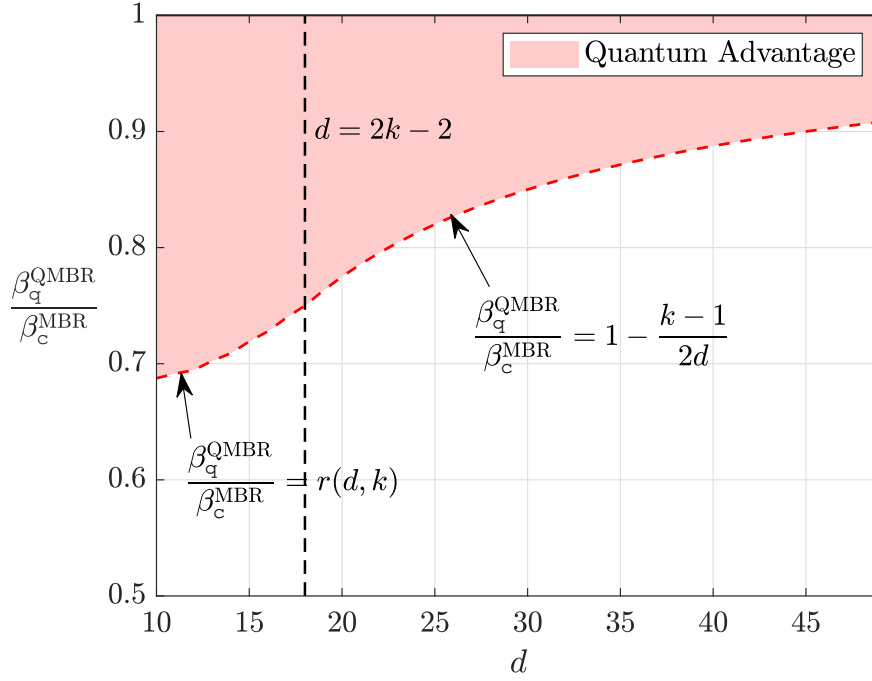


Figure 3: The per-node bandwidth ratio $\frac{\beta_q^{\text{QMBR}}}{\beta_c^{\text{QMBR}}}$ versus the number of helper nodes d , when $n = 50, k = 10$.

$2k - 2$, the ratio increases with d and attains the value $3/4$ at $d = 2k - 2$. For $d \geq 2k - 2$, the ratio continues to increase from $3/4$ and asymptotically approaches 1 as d grows. The curve is illustrated in Fig. 3.

Corollary 3 (Optimal regenerating point) *By comparing Corollary 1 and Corollary 2, we observe that the QMSR and QMBR points coincide when $d \geq 2k - 2$. In this regime, there is a single optimal regenerating point that simultaneously minimizes both the storage cost and the repair bandwidth, thereby breaking the fundamental storage-bandwidth tradeoff present in classical systems. This phenomenon arises because, when $d \geq 2k - 2$, entanglement assistance enables the repair bandwidth at the QMSR point to attain its minimum value, $d\beta_q^{\text{QMSR}} = \alpha^{\text{QMSR}}$, which also achieves the minimization of the bandwidth, i.e., the QMBR point. Consequently, the QMSR and QMBR points coincide.*

5 Illustrative Examples

Example 1 ($d \geq 2k - 2$) *Consider a distributed storage system with parameters $(n, k, d) = (8, 4, 7)$, and normalize the file size $B = 1$. We now compare the storage-bandwidth tradeoff in classical and entanglement-assisted settings. From (14), the classical storage-bandwidth*

tradeoff is given by

$$\sum_{i=0}^3 \min\{(7-i)\beta_c, \alpha\} \geq 1. \quad (39)$$

The corresponding MSR point is thus

$$(\alpha^{\text{MSR}}, d\beta_c^{\text{MSR}}) = \left(\frac{1}{4}, \frac{7}{16}\right), \quad (40)$$

while the MBR point is given by

$$(\alpha^{\text{MBR}}, d\beta_c^{\text{MBR}}) = \left(\frac{7}{22}, \frac{7}{22}\right), \quad (41)$$

where the repair bandwidth equals the per-node storage at the MBR point.

In contrast, under entanglement assistance, Theorem 1 yields the bound

$$\sum_{i=0}^3 \min\{2(7-i)\beta_q, 7\beta_q, \alpha\} = 4 \min\{\alpha, 7\beta_q\} \geq 1, \quad (42)$$

as shown in (17). Hence, the QMSR and QMBR operating points are given by

$$(\alpha^{\text{QMSR}}, d\beta_q^{\text{QMSR}}) = (\alpha^{\text{QMBR}}, d\beta_q^{\text{QMBR}}) = \left(\frac{1}{4}, \frac{1}{4}\right). \quad (43)$$

The classical and quantum curves are shown in Fig. 4. It can be seen that, in the case of minimum storage, we have $\alpha^{\text{QMSR}} = \alpha^{\text{MSR}}$, but the bandwidth is significantly reduced. Indeed, when $d \geq 2k - 2$, entanglement assistance allows the repair bandwidth at the QMSR point to attain its minimum possible value, equal to the per-node storage. This also leads to the minimum bandwidth point. Hence, the QMSR and QMBR points coincide, yielding a single optimal point on the quantum curve.

Example 2 ($d < 2k - 2$) In this example, we focus on the other regime when the number of helper nodes is relatively small, i.e., $d < 2k - 2$. Consider a distributed storage system with parameters $(n, k, d) = (15, 10, 14)$ and a normalized file size $B = 1$. In the classical setting, the storage-bandwidth tradeoff is characterized by

$$\sum_{i=0}^9 \min\{(14-i)\beta_c, \alpha\} \geq 1. \quad (44)$$

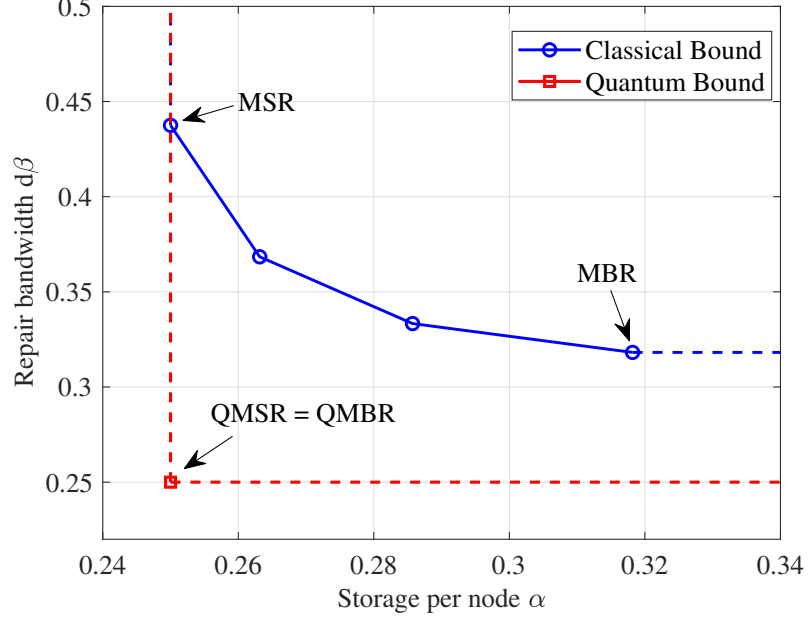


Figure 4: Comparison of the classical and quantum tradeoff for $n = 8$, $k = 4$, $d = 7$, $B = 1$. Since $d \geq 2k - 2$, the QMSR point coincides with the QMBR point, yielding a single optimal regenerating point for the system.

The corresponding MSR and MBR points are given by

$$(\alpha^{\text{MSR}}, d\beta_c^{\text{MSR}}) = \left(\frac{1}{10}, \frac{7}{25} \right), \quad (45)$$

$$(\alpha^{\text{MBR}}, d\beta_c^{\text{MBR}}) = \left(\frac{14}{95}, \frac{14}{95} \right), \quad (46)$$

respectively.

In the quantum setting, the bound in Theorem 1 specializes to

$$\sum_{i=0}^9 \min\{14\beta_q, 2(14-i)\beta_q, \alpha\} \geq 1, \quad (47)$$

which yields the QMSR and QMBR points

$$(\alpha^{\text{QMSR}}, d\beta_q^{\text{QMSR}}) = \left(\frac{1}{10}, \frac{7}{50} \right), \quad (48)$$

$$(\alpha^{\text{QMBR}}, d\beta_q^{\text{QMBR}}) = \left(\frac{7}{67}, \frac{7}{67} \right), \quad (49)$$

respectively. The classical and quantum curves are shown in Fig. 5.

Note that, unlike the regime $d \geq 2k - 2$, there is no single optimal point in the quantum setting. However, we observe that the quantum tradeoff outperforms its classical counterpart across the entire curve. In particular, entanglement-assisted repair achieves a substantial

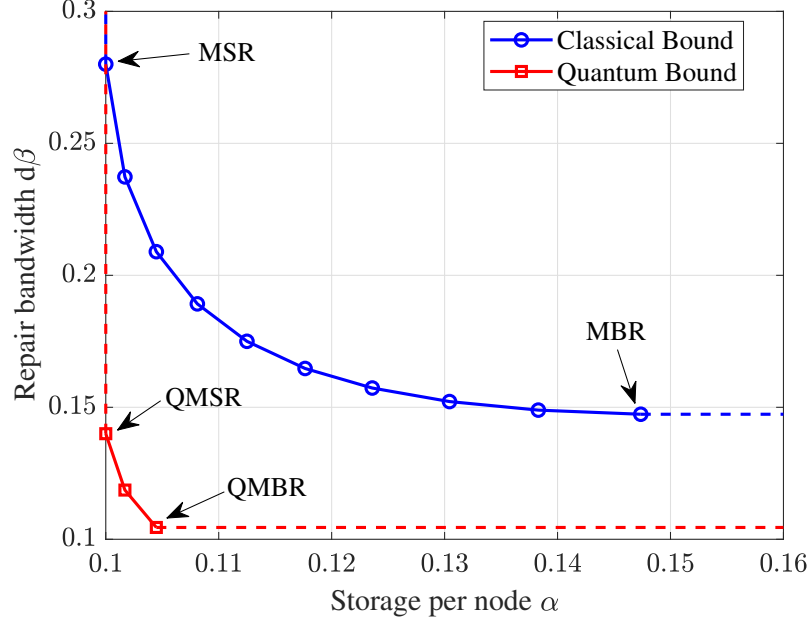


Figure 5: Comparison of the classical and quantum tradeoff curves for $n = 15$, $k = 10$, $d = 14$, and $B = 1$.

reduction in repair bandwidth at the minimum-storage point. Recalling Corollary 2, since $d \leq 2k - 2$ in this example, the repair bandwidth at the QMSR point is reduced by a factor of 2 compared to the classical MSR point.

6 Converse Proof

Consider an (n, k, d) distributed storage system over $\mathbb{F}_q$. Each node i stores a classical random variable $W_i \in \mathbb{F}_q^\alpha$. From the data retrieval property, we have

$$H(W_{\mathcal{K}}) = B, \quad \forall \mathcal{K} \subseteq [n] \text{ with } |\mathcal{K}| = k, \quad (50)$$

where $H(\cdot)$ is the Shannon entropy.

When a node f fails, let $\mathcal{D} \subset [n] \setminus \{f\}$ be any helper set with $|\mathcal{D}| = d$. Each helper $i \in \mathcal{D}$ sends a quantum system $\mathcal{Q}_i \triangleq \mathcal{Q}_{Q_i}$ to the newcomer with $\log_q |\mathcal{Q}_i| = \beta_{\mathbf{q}}$, i.e., $\beta_{\mathbf{q}}$ qudits are sent by each helper. The newcomer generates $W_f \in \mathbb{F}_q^\alpha$ by measuring the received quantum system such that the data retrieval property continues to be satisfied.

We now specify the state of the whole system in the repair process. Consider that the helper nodes in $\mathcal{D}$ have classical information described by $\sum_{w_{\mathcal{D}}} \Pr(W_{\mathcal{D}} = w_{\mathcal{D}}) |w_{\mathcal{D}}\rangle \langle w_{\mathcal{D}}|_{\mathcal{Q}_1}$ and two replicated initial quantum systems $\mathcal{Q}_2$ and $\mathcal{Q}_3$, described by $\rho_{\mathcal{Q}_2}^{\text{ini}}$ and $\rho_{\mathcal{Q}_3}^{\text{ini}}$, respectively. The initial quantum systems are independent with the classical storage. Hence, the

density matrix of the initial system is given by

$$\rho^{\text{ini}} = \sum_{w_{\mathcal{D}}, w_f} \Pr(W_{\mathcal{D}} = w_{\mathcal{D}}, W_f = w_f) \left(|w_{\mathcal{D}}\rangle \langle w_{\mathcal{D}}|_{Q_1} \otimes \rho_{Q_2}^{\text{ini}} \otimes \rho_{Q_3}^{\text{ini}} \otimes |w_f\rangle \langle w_f|_{Q_4} \right). \quad (51)$$

After each node in $\mathcal{D}$ encodes classical information into Q_2 and Q_3 , the quantum state becomes

$$\rho^{\text{enc}} = \sum_{w_{\mathcal{D}}, w_f} \Pr(W_{\mathcal{D}} = w_{\mathcal{D}}, W_f = w_f) \left(|w_{\mathcal{D}}\rangle \langle w_{\mathcal{D}}|_{Q_1} \otimes \rho_{Q_2}^{\text{enc}} \otimes \rho_{Q_3}^{\text{enc}} \otimes |w_f\rangle \langle w_f|_{Q_4} \right). \quad (52)$$

The helper nodes then send the quantum system Q_3 and keep the Q_2 . The newcomer measures the received quantum system, resulting in a quantum system as

$$\begin{aligned} \bar{\rho}^{\text{enc}} = \sum_{w_{\mathcal{D}}, w_f, z_f} \Pr(W_{\mathcal{D}} = w_{\mathcal{D}}, W_f = w_f, Z_f = z_f) \\ \times \left(|w_{\mathcal{D}}\rangle \langle w_{\mathcal{D}}|_{Q_1} \otimes \rho_{Q_2}^{\text{enc}} \otimes |z_f\rangle \langle z_f|_{Q_3} \otimes |w_f\rangle \langle w_f|_{Q_4} \right), \end{aligned} \quad (53)$$

where Z_f is the measurement result.

By data processing inequality [22], we have

$$S(W_f|Q_3)_{\rho^{\text{enc}}} \leq S(W_f|Z_f)_{\bar{\rho}^{\text{enc}}} = 0, \quad (54)$$

where $S(\cdot)$ denotes the von Neumann entropy, and the last equality is because the storage W_f is generated from the measurement result Z_f .

Now, fix any subset $\mathcal{A} \subseteq \mathcal{D}$. Since $S(W_f|Q_{\mathcal{D}})_{\rho^{\text{enc}}} \leq 0$, we have

$$S(W_f|Q_{\mathcal{A}})_{\rho^{\text{enc}}} \leq I(W_f; Q_{\mathcal{D} \setminus \mathcal{A}}|Q_{\mathcal{A}})_{\rho^{\text{enc}}}. \quad (55)$$

We now need the following lemma to bound $I(W_f; Q_{\mathcal{D} \setminus \mathcal{A}}|Q_{\mathcal{A}})_{\rho^{\text{enc}}}$.

Lemma 1 (Dimension bound with quantum side information) *Let U be a classical random variable and let Q_A, Q_B be finite-dimensional quantum systems. For any state ρ_{UAB} ,*

$$I(U; B|A)_{\rho_{UAB}} \leq \min \{ 2 \log_q |Q_B|, \log_q |Q_B Q_A| \}, \quad (56)$$

where

$$\rho_{UAB} = \sum_u p_u |u\rangle \langle u|_U \otimes \rho_{AB}^u. \quad (57)$$

Proof: First, using von Neumann entropies,

$$I(U; B|A) = S(B|A) - S(B|UA). \quad (58)$$

Then, $S(B|A) \leq S(B) \leq \log_q |\mathcal{Q}_B|$, and by the Araki-Lieb inequality [23]

$$|S(AU) - S(B)| \leq S(AUB), \quad (59)$$

we obtain

$$S(AUB) \geq S(AU) - S(B). \quad (60)$$

Hence,

$$S(B|AU) \geq -S(B) \geq -\log_q |\mathcal{Q}_B|. \quad (61)$$

Therefore,

$$I(U; B|A) \leq 2 \log_q |\mathcal{Q}_B|. \quad (62)$$

On the other hand,

$$I(U; B|A) = I(U; BA) - I(U; A) \quad (63)$$

$$\leq I(U; BA) \leq \log_q |\mathcal{Q}_B \mathcal{Q}_A|. \quad (64)$$

This completes the proof. ■

By applying Lemma 1 to (55), we have

$$S(W_f|Q_{\mathcal{A}})_{\rho^{\text{enc}}} \leq \min\{2(d - |\mathcal{A}|)\beta_{\mathbf{q}}, d\beta_{\mathbf{q}}\}. \quad (65)$$

Moreover, since we have

$$S(W_f|Q_{\mathcal{A}})_{\rho^{\text{enc}}} \leq S(W_f) \leq \alpha, \quad (66)$$

$S(W_f|Q_{\mathcal{A}})_{\rho^{\text{enc}}}$ is thus bounded by

$$S(W_f|Q_{\mathcal{A}})_{\rho^{\text{enc}}} \leq \min\{\alpha, 2(d - |\mathcal{A}|)\beta_{\mathbf{q}}, d\beta_{\mathbf{q}}\}. \quad (67)$$

Now, from the independence between the initial quantum system and storage, we have

$$I(W_f; W_{\mathcal{A}})_{\rho^{\text{ini}}} = I(W_f; W_{\mathcal{A}} Q_3)_{\rho^{\text{ini}}} \quad (68)$$

$$\geq I(W_f; Q_3)_{\rho^{\text{enc}}} \geq I(W_f; Q_{\mathcal{A}})_{\rho^{\text{enc}}}. \quad (69)$$

Hence, we have

$$I(W_f; W_{\mathcal{A}})_{\rho^{\text{ini}}} \geq I(W_f; Q_{\mathcal{A}})_{\rho^{\text{enc}}} \quad \Rightarrow \quad S(W_f|W_{\mathcal{A}})_{\rho^{\text{ini}}} \leq S(W_f|Q_{\mathcal{A}})_{\rho^{\text{enc}}}. \quad (70)$$

From (67) and (70), we have

$$S(W_f|W_{\mathcal{A}})_{\rho^{\text{ini}}} \leq \min\{\alpha, 2(d - |\mathcal{A}|)\beta_{\mathbf{q}}, d\beta_{\mathbf{q}}\}. \quad (71)$$

Finally, using the chain rule for joint entropy

$$B = H(W_{[k]}) = \sum_{i=1}^k H(W_i|W_{[i-1]}). \quad (72)$$

For each i , applying (71) with $f = i$ and choosing $\mathcal{A} = [i - 1]$. This is valid because $d \geq k$ implies that we can select a helper set $\mathcal{D}$ of size d that contains $[i - 1]$, i.e., $\mathcal{A} = [i - 1] \subseteq \mathcal{D}$. Then, $|\mathcal{A}| = i - 1$ and (71) yields

$$B \leq \sum_{i=1}^k \min\{2(d - i + 1)\beta_{\mathbf{q}}, d\beta_{\mathbf{q}}, \alpha\} \quad (73)$$

$$= \sum_{i=0}^{k-1} \min\{2(d - i)\beta_{\mathbf{q}}, d\beta_{\mathbf{q}}, \alpha\}. \quad (74)$$

This completes the converse proof of Theorem 1.

7 Achievability Proof

In this section, we establish the achievability of the proposed storage-bandwidth tradeoff by extending the randomized code construction framework of [24]. We first prove the existence of coding schemes that achieve the QMSR point while preserving the data retrieval property. We then generalize the construction to arbitrary points on the storage-bandwidth tradeoff curve. Notably, achieving the entire tradeoff region requires maintaining a strictly stronger invariant than the one used at the QMSR point. This strengthened invariant and the corresponding scheme will be developed in Section 7.2.

A key tool in the existence proof is the Schwartz–Zippel lemma, which guarantees the existence of suitable coding coefficients over sufficiently large finite fields via a probabilistic argument.

Lemma 2 (Schwartz–Zippel lemma [25]) *Let $Q(x_1, \dots, x_n)$ be a nonzero multivariate polynomial over a finite field $\mathbb{F}$ with total degree at most d_0 . If the variables $x_1, \dots, x_n$ are chosen independently and uniformly from $\mathbb{F}$, then*

$$\Pr(Q(x_1, \dots, x_n) = 0) \leq \frac{d_0}{|\mathbb{F}|}, \quad (75)$$

where $|\mathbb{F}|$ denotes the field size.

As a consequence, for any nonzero polynomial Q , if $|\mathbb{F}| > d_0$, then there exists at least one assignment of the variables such that $Q(x_1, \dots, x_n) \neq 0$.

7.1 Achievability Proof of the QMSR Point

We establish achievability of the QMSR point via induction on the number of node failures and subsequent repairs. The central idea is to preserve the data retrieval property as a global invariant under an arbitrary sequence of repair operations.

Theorem 2 *Consider an $(n, k, d, \alpha, \beta_{\mathbf{q}}, B)$ distributed storage system at the QMSR point. Over a finite field $\mathbb{F}_q$ with*

$$q > d_0 \triangleq \binom{n}{k} B_0, \quad (76)$$

there exists a linear coding scheme such that the data retrieval property is preserved under any sequence of node failures and repairs.

The proof of Theorem 2 proceeds by induction, as detailed below.

7.1.1 Invariant Description

In the proposed construction, the file is divided into two independent instances, each of size

$$B_0 = k\alpha_0, \quad \alpha_0 \triangleq \min\{d, 2(d - k + 1)\}. \quad (77)$$

Thus, the total file size is $B = 2B_0$. At the QMSR point, the per-helper repair bandwidth satisfies $\beta_{\mathbf{q}} = 2$.

We employ linear coding for storage. Let the overall file be represented by $\mathbf{x} \in \mathbb{F}_q^{B \times 1}$. Node i stores $\alpha = 2\alpha_0$ symbols of the form

$$\mathbf{x}^T \mathbf{Q}_i, \quad i \in [n], \quad (78)$$

where the storage matrix has the block-diagonal structure

$$\mathbf{Q}_i = \begin{bmatrix} \bar{\mathbf{Q}}_i & \mathbf{0}_{B_0 \times \alpha_0} \\ \mathbf{0}_{B_0 \times \alpha_0} & \bar{\mathbf{Q}}_i \end{bmatrix}, \quad (79)$$

where $\bar{\mathbf{Q}}_i \in \mathbb{F}_q^{B_0 \times \alpha_0}$ denotes the coding matrix associated with each instance of the file.

The data retrieval property requires that for any index set $s_{[k]} \subseteq [n]$,

$$\text{rank}([\mathbf{Q}_{s_1}, \dots, \mathbf{Q}_{s_k}]) = B. \quad (80)$$

Due to the block-diagonal structure of $\mathbf{Q}_i$, this condition is equivalent to

$$\text{rank}([\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_k}]) = B_0, \quad (81)$$

i.e., the data retrieval property holds for each instance of the file. Our objective is to show that the rank condition in (81) can be maintained after every repair process, in which each newcomer downloads $\beta_q = 2$ qudits from each of the d helper nodes. The remainder of the proof establishes that this rank invariant can be maintained inductively for an arbitrary sequence of node failures and repairs.

7.1.2 Induction Process

We begin by verifying that the data retrieval property (81) holds at the initial stage. Let the concatenated matrix $[\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n]$ be a collection of $B_0 \times n\alpha_0$ independent variables over $\mathbb{F}_q$. For each k -subset $s_{[k]} \subseteq [n]$, define the polynomial

$$f_{s_{[k]}} \triangleq \det([\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_k}]). \quad (82)$$

Each $f_{s_{[k]}}$ is a nonzero multivariate polynomial, since the matrix $[\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_k}]$ can be assigned to any invertible $B_0 \times B_0$ matrix so that the determinant polynomial $f_{s_{[k]}}$ is not identically zero.

Now, consider the product polynomial

$$f \triangleq \prod_{s_{[k]} \subseteq [n]} f_{s_{[k]}} = \prod_{s_{[k]} \subseteq [n]} \det([\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_k}]). \quad (83)$$

This polynomial is nonzero and has total degree at most $d_0 = \binom{n}{k} B_0$. This is because f is the product of $\binom{n}{k}$ determinant polynomials, each corresponding to a $B_0 \times B_0$ submatrix and hence having degree B_0 . (The determinant of an $m \times m$ matrix $\mathbf{M}$ is a polynomial of degree at most m [24].) By the Schwartz–Zippel lemma, if the field size q satisfies $q > d_0$, then there exists an assignment of $[\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n]$ such that $f \neq 0$, i.e., all determinants, $\det([\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_k}]), s_{[k]} \subseteq [n]$, are simultaneously nonzero. Therefore, an initial storage configuration satisfying the data retrieval property exists, which completes the base case of induction.

Now, assume that the data retrieval property holds after the m th repair. We show that it can be preserved after the $(m+1)$ th repair. Without loss of generality, suppose node 1 fails, and the newcomer connects to any helper nodes $s'_{[d]} \subseteq [n] \setminus \{1\}$. Each helper sends $\beta_q = 2$ qudits. The newcomer performs quantum measurements followed by linear post-processing to generate its stored α symbols. The repair process is specified as follows.

1. **Entanglement distribution:** The d helper nodes share an entangled state

$$|\psi\rangle \in \text{CSS}(\text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{u}), \text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{v})), \quad (84)$$

where $\boldsymbol{\alpha}, \mathbf{u}, \mathbf{v} \in \mathbb{F}_q^{2d \times 1}$ are GRS parameters chosen such that $\text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{u})$ and $\text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{v})$ are dual. Thus, the corresponding initial density matrix is described by

$$\rho^{\text{ini}} = |\psi\rangle \langle \psi|. \quad (85)$$

2. **Information encoding:** Let $\mathbf{x}(1)$ and $\mathbf{x}(2) \in \mathbb{F}_q^{B_0 \times 1}$ denote the two instances of the file. Each node s'_i , $i \in [d]$, applies a local precoding map to its storage as

$$(\mathbf{x}(1))^T \bar{\mathbf{Q}}_{s'_i} \mathbf{V}_{s'_i} \text{diag}(u_{2i-1}, u_{2i})^{-1}, \quad (\mathbf{x}(2))^T \bar{\mathbf{Q}}_{s'_i} \mathbf{V}_{s'_i} \text{diag}(v_{2i-1}, v_{2i})^{-1}, \quad (86)$$

where $\mathbf{V}_{s'_i}, i \in [d]$, is the linear precoding matrix to be designed. Define $\bar{\mathbf{P}}_u = [\bar{\mathbf{Q}}_{s'_1} \mathbf{V}_{s'_1}, \dots, \bar{\mathbf{Q}}_{s'_d} \mathbf{V}_{s'_d}] \text{diag}(\mathbf{u})^{-1}$ and $\bar{\mathbf{P}}_v = [\bar{\mathbf{Q}}_{s'_1} \mathbf{V}_{s'_1}, \dots, \bar{\mathbf{Q}}_{s'_d} \mathbf{V}_{s'_d}] \text{diag}(\mathbf{v})^{-1}$. The classical information embedded in the Pauli parameters is then

$$\mathbf{x}_u^T \triangleq (\mathbf{x}(1))^T \bar{\mathbf{P}}_u, \quad \mathbf{x}_v^T \triangleq (\mathbf{x}(2))^T \bar{\mathbf{P}}_v. \quad (87)$$

Node $s'_i, i \in [d]$, applies $\mathbf{X}(\mathbf{x}_u(i))\mathbf{Z}(\mathbf{x}_v(i))$ to its local quantum system. The encoded quantum state is then described by

$$\rho^{\text{enc}} = (\mathbf{X}(\mathbf{x}_u)\mathbf{Z}(\mathbf{x}_v)) \rho^{\text{ini}} (\mathbf{X}(\mathbf{x}_u)\mathbf{Z}(\mathbf{x}_v))^{\dagger}. \quad (88)$$

3. **Storage generation:** Upon receiving the encoded quantum system, the newcomer performs the syndrome measurement and obtains

$$\mathbf{s}_X = (\text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{u}))^T \mathbf{x}_u, \quad \mathbf{s}_Z = (\text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{v}))^T \mathbf{x}_v. \quad (89)$$

Equivalently, stacking the syndromes gives

$$\begin{aligned} [\mathbf{s}_X^T, \mathbf{s}_Z^T] &= \mathbf{x}^T \begin{bmatrix} \bar{\mathbf{P}}_u \cdot \text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{u}) & \mathbf{0}_{B_0 \times d} \\ \mathbf{0}_{B_0 \times d} & \bar{\mathbf{P}}_v \cdot \text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{v}) \end{bmatrix} \\ &= \mathbf{x}^T \begin{bmatrix} \bar{\mathbf{Q}} \cdot \text{diag}(\mathbf{u})^{-1} \cdot \text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{u}) & \mathbf{0}_{B_0 \times d} \\ \mathbf{0}_{B_0 \times d} & \bar{\mathbf{Q}} \cdot \text{diag}(\mathbf{v})^{-1} \cdot \text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{v}) \end{bmatrix}, \end{aligned} \quad (90)$$

where $\bar{\mathbf{Q}} \triangleq [\bar{\mathbf{Q}}_{s'_1} \mathbf{V}_{s'_1}, \dots, \bar{\mathbf{Q}}_{s'_d} \mathbf{V}_{s'_d}]$. Note that $\text{GRS}_{d,2d}^q(\boldsymbol{\alpha}, \mathbf{u}) = \text{diag}(\mathbf{u})\mathbf{A}$, where $\mathbf{A}$ is a

Vandermonde matrix as shown in (5). Hence, (90) reduces to

$$[\mathbf{s}_X^T, \mathbf{s}_Z^T] = \mathbf{x}^T \begin{bmatrix} \bar{\mathbf{Q}}\mathbf{A} & \mathbf{0}_{B_0 \times d} \\ \mathbf{0}_{B_0 \times d} & \bar{\mathbf{Q}}\mathbf{A} \end{bmatrix}. \quad (91)$$

The newcomer then applies a linear post-processing matrix in the form

$$\mathbf{V}_{\text{dec}} = \begin{bmatrix} \bar{\mathbf{V}}_{\text{dec}} & \mathbf{0}_{d \times \alpha_0} \\ \mathbf{0}_{d \times \alpha_0} & \bar{\mathbf{V}}_{\text{dec}} \end{bmatrix}, \quad (92)$$

where $\bar{\mathbf{V}}_{\text{dec}} \in \mathbb{F}_q^{d \times \alpha_0}$ is the decoding matrix for each instance. Consequently, the storage matrix at the repaired node has the form

$$\mathbf{Q}'_1 \triangleq \begin{bmatrix} \bar{\mathbf{Q}}'_1 & \mathbf{0}_{B_0 \times \alpha_0} \\ \mathbf{0}_{B_0 \times \alpha_0} & \bar{\mathbf{Q}}'_1 \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{V}}_{\text{dec}} & \mathbf{0}_{B_0 \times \alpha_0} \\ \mathbf{0}_{B_0 \times \alpha_0} & \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{V}}_{\text{dec}} \end{bmatrix} \in \mathbb{F}_q^{B \times \alpha}. \quad (93)$$

Remark 5 *Note that in the scheme, we adopt a GRS-based CSS codes to prepare the shared entangled state and perform syndrome measurement for two main reasons. First, syndrome measurement produces linear combinations of the encoded classical parameters. Since the newcomer ultimately stores linear combinations of the original file symbols, this linear structure is essential. Second, the CSS construction based on dual GRS codes provides a parity-check matrix with sufficiently large rank. This guarantees that the useful classical information embedded by the helper nodes can be transferred to the newcomer without rank loss. In particular, the Vandermonde structure of the underlying GRS matrix ensures that appropriately selected columns remain the maximum rank, which is crucial for preserving the data retrieval invariant.*

Now, to ensure the data retrieval property for each instance, it suffices to require that for every index set $s_{[k-1]} \subseteq [2 : n]$

$$\det([\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]) \neq 0, \quad (94)$$

i.e., the file can be retrieved from the $\bar{\mathbf{Q}}_1$ at the newcomer and the storage of any other $k-1$ nodes. Hence, we define the product polynomial

$$P \triangleq \prod_{s_{[k-1]} \subseteq [2:n]} \det([\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]), \quad (95)$$

which is a multivariate polynomial in the entries of $\bar{\mathbf{V}}_{s'_{[d]}}$ and $\bar{\mathbf{V}}_{\text{dec}}$. The degree of P is at most $2\binom{n-1}{k-1}\alpha_0$ as each of the $\binom{n-1}{k-1}$ determinant terms has degree at most $2\alpha_0$. Note that $2\binom{n-1}{k-1}\alpha_0 \geq d_0$ for $n \geq 2$. To apply the Schwartz–Zippel lemma, it suffices to show that P is not identically zero. This follows if each determinant term in (95) is itself a nonzero

polynomial. We now show that for each determinant term, one can choose the variables such that it evaluates to a nonzero value.

To this end, we fix an arbitrary subset $s_{[k-1]}$ and show that there exists a choice of the precoding matrices $\bar{\mathbf{V}}_{s'_{[d]}}$ and decoding matrix $\bar{\mathbf{V}}_{\text{dec}}$ such that $\det([\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]) \neq 0$. The key ingredient is the following column-selection lemma.

Lemma 3 (Column-selection lemma) *Let $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$ be matrices over $\mathbb{F}_q$, where $\bar{\mathbf{Q}}_i \in \mathbb{F}_q^{k\alpha_0 \times \alpha_0}$ and $\alpha_0 \triangleq \min\{2(d-k+1), d\}$.⁴ Assume that the data retrieval property holds, i.e.,*

$$\text{rank}([\bar{\mathbf{Q}}_{t_1}, \dots, \bar{\mathbf{Q}}_{t_k}]) = k\alpha_0, \quad \forall t_{[k]} \subseteq [n]. \quad (96)$$

Fix any subset $s_{[k-1]} \subseteq [n]$ and any subset $s'_{[d]} \subseteq [n]$. Then, there exist index sets $\mathcal{I}_j \subseteq [\alpha_0]$, $j \in s'_{[d]} \setminus s_{[k-1]}$ with $|\mathcal{I}_j| \leq 2$, and

$$\sum_{j \in s'_{[d]} \setminus s_{[k-1]}} |\mathcal{I}_j| = \alpha_0, \quad (97)$$

such that the matrix formed by the α_0 selected columns

$$\mathbf{C} \triangleq [\bar{\mathbf{Q}}_j(:, \mathcal{I}_j) : j \in s'_{[d]} \setminus s_{[k-1]}] \in \mathbb{F}_q^{k\alpha_0 \times \alpha_0} \quad (98)$$

satisfies

$$\text{rank}([\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}, \mathbf{C}]) = k\alpha_0. \quad (99)$$

Equivalently, the selected α_0 columns are linearly independent with $[\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]$.

Remark 6 *Lemma 3 states that for any fixed $k-1$ storage $\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}$, we can extract exactly α_0 columns from the helper nodes in $s'_{[d]} \setminus s_{[k-1]}$, using at most two columns from each helper. These α_0 columns are linearly independent together with $[\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]$, thus the total rank becomes $k\alpha_0$. Therefore, we can design the precoding matrices $\bar{\mathbf{V}}_{s'_{[d]}}$ so that the newcomer stores precisely these α_0 independent columns. With this choice, the storage matrix at the newcomer, $\bar{\mathbf{Q}}'_1$, satisfies $\det([\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]) \neq 0$, i.e., each determinant term of P in (95) is a nonzero polynomial.*

Proof: For each node index $i \in [n]$, let us denote

$$\bar{\mathbf{Q}}_i = [\bar{\mathbf{q}}_{i,1}, \dots, \bar{\mathbf{q}}_{i,\alpha_0}], \quad (100)$$

⁴We choose α_0 with this value to make sure that β_0 is an integer in the achievable scheme.

where $\bar{\mathbf{Q}}_{i,j} \in \mathbb{F}_q^{k\alpha_0 \times 1}$, $j \in [\alpha_0]$. Fix any index set $s_{[k-1]} \subseteq [n]$ and define

$$\bar{\mathbf{Q}}^{(k-1)} \triangleq [\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}], \quad (101)$$

and $\mathcal{W} \triangleq \text{colspan}(\bar{\mathbf{Q}}^{(k-1)})$ as a column space. By the data retrieval property, for any $s'_j \in s'_{[d]} \setminus s_{[k-1]}$, the matrix $[\bar{\mathbf{Q}}^{(k-1)}, \bar{\mathbf{Q}}_{s'_j}]$ has rank $k\alpha_0$. Since $\bar{\mathbf{Q}}_{s'_j}$ has α_0 columns, we have

$$\text{rank}(\bar{\mathbf{Q}}^{(k-1)}) = (k-1)\alpha_0. \quad (102)$$

Now, consider the quotient space $\mathcal{V} = \mathbb{F}_q^{k\alpha_0} / \mathcal{W}$, which has dimension

$$\dim(\mathcal{V}) = k\alpha_0 - \dim(\mathcal{W}) = k\alpha_0 - (k-1)\alpha_0 = \alpha_0. \quad (103)$$

Let $\pi : \mathbb{F}_q^{k\alpha_0} \rightarrow \mathcal{V}$ be the canonical projection. Again by the data retrieval property, for each $s'_j \in s'_{[d]} \setminus s_{[k-1]}$, the projected columns $\{\pi(\bar{\mathbf{q}}_{s'_j,1}), \dots, \pi(\bar{\mathbf{q}}_{s'_j,\alpha_0})\}$ span $\mathcal{V}$.

Let $r \triangleq d - k + 1$. Since $\alpha_0 = \min\{2r, d\} \leq 2r$, we may select vectors greedily from $\bar{\mathbf{Q}}_{s'_{[d]} \setminus s_{[k-1]}}$ as follows: At each step, choose at most two columns from the current block whose projections are linearly independent of those previously chosen. Because *each block spans* $\mathcal{V}$, this procedure increases the span dimension until α_0 independent vectors in $\mathcal{V}$ are obtained, using at most two columns from each $\bar{\mathbf{Q}}_{s'_j}$. Together with the $(k-1)\alpha_0$ columns selected from $\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}$, they form $k\alpha_0$ linearly independent vectors. This completes the proof. ■

By Lemma 3, for any fixed $\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}$, we can select exactly α_0 columns from helper nodes $s'_{[d]} \setminus s_{[k-1]}$ to generate the newcomer's storage, using at most two columns from each helper, such that together with $\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}$, the total rank is $k\alpha_0$. For each $s'_j \in s'_{[d]} \setminus s_{[k-1]}$, let $r_j \triangleq |\mathcal{I}_j| \leq 2$, and write $\mathcal{I}_j$ as

$$\mathcal{I}_j \triangleq \{\mathcal{I}_j(1), \dots, \mathcal{I}_j(r_j)\}. \quad (104)$$

We design the precoding matrices as column-selection matrices,

$$\bar{\mathbf{V}}_j = [\mathbf{e}_{\mathcal{I}_j(1)}, \dots, \mathbf{e}_{\mathcal{I}_j(r_j)}, \mathbf{0}_{\alpha_0 \times (2-r_j)}] \in \mathbb{F}_q^{\alpha_0 \times 2}, \quad s'_j \in s'_{[d]} \setminus s_{[k-1]}, \quad (105)$$

where $\mathbf{e}_\ell \in \mathbb{F}_q^{\alpha_0 \times 1}$ denotes the ℓ th standard basis vector. For helper nodes $j \in s'_{[d]} \cap s_{[k-1]}$, we set $\bar{\mathbf{V}}_j = \mathbf{0}_{\alpha_0 \times 2}$. With this choice, we have

$$\text{rank}([\bar{\mathbf{Q}}_{s'_1} \mathbf{V}_{s'_1}, \dots, \bar{\mathbf{Q}}_{s'_d} \mathbf{V}_{s'_d}, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]) = k\alpha_0. \quad (106)$$

Let $\mathcal{S}$ denote the indices of the α_0 nonzero columns in $\bar{\mathbf{Q}} = [\bar{\mathbf{Q}}_{s'_1} \mathbf{V}_{s'_1}, \dots, \bar{\mathbf{Q}}_{s'_d} \mathbf{V}_{s'_d}]$, and $\tilde{\mathbf{Q}} \triangleq \bar{\mathbf{Q}}_{[:,\mathcal{S}]} \in \mathbb{F}_q^{k\alpha_0 \times \alpha_0}$ be the submatrix formed by these columns. By this construction, we

have

$$\text{rank} \left([\bar{\mathbf{Q}}, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}] \right) = k\alpha_0. \quad (107)$$

Recall from (93) that the newcomer's storage matrix for each instance is given by

$$\bar{\mathbf{Q}}'_1 = \bar{\mathbf{Q}} \mathbf{A} \bar{\mathbf{V}}_{\text{dec}}. \quad (108)$$

By construction, only the columns of $\bar{\mathbf{Q}}$ indexed by $\mathcal{S}$ are nonzero. Let $\tilde{\mathbf{A}} \triangleq \mathbf{A}_{[\mathcal{S},:] } \in \mathbb{F}_q^{\alpha_0 \times d}$ denote the submatrix of $\mathbf{A}$ formed by the rows indexed by $\mathcal{S}$. Since the remaining columns of $\bar{\mathbf{Q}}$ are zero, we have

$$\bar{\mathbf{Q}} \mathbf{A} = \tilde{\mathbf{Q}} \tilde{\mathbf{A}}. \quad (109)$$

We therefore choose the decoding matrix as

$$\bar{\mathbf{V}}_{\text{dec}} = \tilde{\mathbf{A}}^T. \quad (110)$$

With this design, the repaired node's storage matrix for each instance is

$$\bar{\mathbf{Q}}'_1 = \tilde{\mathbf{Q}} \tilde{\mathbf{A}} \tilde{\mathbf{A}}^T. \quad (111)$$

Also, $\tilde{\mathbf{A}} \tilde{\mathbf{A}}^T$ is a $\alpha_0 \times \alpha_0$ full-rank matrix, which preserves the subspace $\text{colspan}(\tilde{\mathbf{Q}})$. Hence, for each instance of the file, the storage matrix $\bar{\mathbf{Q}}'_1$ at the newcomer together with $\bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}$ satisfy the data retrieval property, i.e.,

$$\det([\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_{s_1}, \dots, \bar{\mathbf{Q}}_{s_{k-1}}]) \neq 0. \quad (112)$$

This means that there exists an assignment of the variables such that each determinant term of P is nonzero, and thus, P is a nonzero polynomial.

Now, by the Schwartz-Zippel lemma, if the file size is larger than d_0 in (76), then there exists a common assignment of $\mathbf{V}_{s'_{[d]}}$ and $\mathbf{V}_{\text{dec}}$ such that the data retrieval property holds after the $(m+1)$ th repair. By induction, it holds for all repair rounds. This completes the achievability proof at the QMSR point.

7.2 Achievability Proof for the General Case

We now establish the achievability of the entire storage-bandwidth tradeoff characterized in Theorem 1. A key difficulty, compared to the QMSR point, is that the data retrieval property alone is not sufficient to guarantee that a code attains the general cut-set bound. The following example illustrates this limitation.

Example 3 As shown in Fig. 6, consider a distributed storage system with parameters $n = 4$, $k = 2$, $d = 3$, $\alpha = 4$, and $\beta_q = 1$. From Theorem 1, the cut-set bound gives the maximum file size

$$B = \sum_{i=0}^1 \min\{2(3-i), 3, 4\} = 3 + 3 = 6. \quad (113)$$

We now construct a code that satisfies the data retrieval property initially, but fails to sustain $B = 6$ after one repair.

Let the file consist of 6 symbols $x_1, \dots, x_6$ over a finite field $\mathbb{F}_q$. Suppose node 1 fails and the newcomer connects to the helper nodes 2, 3, 4. The storage at the surviving nodes is:

- **Node 2:** $(x_1, x_2, 0, 0)$;
- **Node 3:** (x_3, x_4, x_5, x_6) ;
- **Node 4:** $(x_1 + x_3, x_2 + x_4, x_1 + x_5, x_2 + x_6)$.

It can be verified that the data retrieval property holds initially, i.e., any two nodes contain enough independent linear combinations to recover all six file symbols.

Now, consider the repair of node 1. Since $\beta_q = 1$, each helper transmits one qudit, thus the newcomer receives a total of $d\beta_q = 3$ qudits. After measurement and classical post-processing, the repaired node stores $\alpha = 4$ classical symbols. By the Holevo bound, 3 qudits can convey at most 3 dits of classical information. Now, consider a DC that connects to node 2 and the repaired node (say node 5). Since node 2 only has two independent symbols and node 5 stores at most three independent symbols, the DC has access to at most 5 independent symbols. Thus, the file with 6 independent symbols cannot be reconstructed after the repair. The data retrieval property is thus violated.

This example demonstrates that the data retrieval property alone is *insufficient* to guarantee the feasibility of the cut-set bound, and that additional structural constraints on storage are necessary.

At the QMSR point, the data retrieval property effectively forces every stored symbol at each node to carry independent information, since the storage dimension is tight and any redundancy would violate the full-rank requirement. However, for the other points on the tradeoff curve, the data retrieval property alone only guarantees that some selection of stored symbols yields full-rank. It does not prevent certain stored symbols from being linearly dependent or even inactive (for instance, some nodes may store 0s). As a consequence, although the data retrieval property may hold before repair, the limited repair bandwidth may be insufficient to regenerate enough independent symbols after one repair. Therefore, for the general case, a stronger invariant than the data retrieval property is necessary.

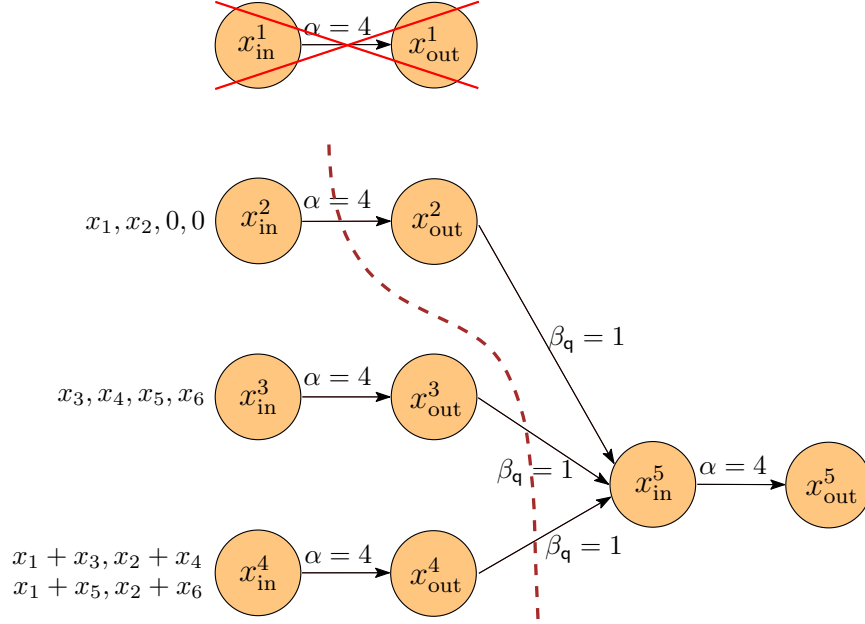


Figure 6: A distributed storage system with $(n, k, d) = (4, 2, 3)$, $\alpha = 4$, $\beta_q = 1$, in which the data retrieval property is satisfied in the initial stage, but the data retrieval property does not hold after the repair.

7.2.1 Invariant Description

We consider two independent instances of the file, and the storage matrix at each node is given by

$$\mathbf{Q}_i = \begin{bmatrix} \bar{\mathbf{Q}}_i & \mathbf{0}_{B_0 \times \alpha_0} \\ \mathbf{0}_{B_0 \times \alpha_0} & \bar{\mathbf{Q}}_i \end{bmatrix}, \quad i \in [n]. \quad (114)$$

The invariant is defined on the storage code $\bar{\mathbf{Q}}_{[n]}$. First, define c_i as

$$c_i = \min\{2(d - i + 1)\beta_q, d\beta_q, \alpha\}, \quad i \in [k], \quad (115)$$

and $\mathbf{c}$ as the length- n vector $\mathbf{c} \triangleq [c_1, \dots, c_k, 0, \dots, 0]$. The invariant is then based on a rank test for the code $\bar{\mathbf{Q}}_{[n]}$ with respect to certain vectors.

Definition 6 A code $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$ is said to pass the rank test by $\mathbf{h} = [h_1, \dots, h_n] \in \mathbb{Z}_+^n$ with $h_i \in [0 : \alpha_0]$, $\sum_{i=1}^n h_i = B_0$, if

$$\text{rank}([\bar{\mathbf{Q}}_1 \mathbf{E}_{h_1}, \dots, \bar{\mathbf{Q}}_n \mathbf{E}_{h_n}]) = B_0, \quad (116)$$

where $\bar{\mathbf{Q}}_i \mathbf{E}_{h_i}$ is the submatrix by choosing the first h_i columns of $\bar{\mathbf{Q}}_i$, i.e.,

$$\mathbf{E}_{h_i} \triangleq \begin{bmatrix} \mathbf{I}_{h_i} \\ \mathbf{0}_{(\alpha_0 - h_i) \times h_i} \end{bmatrix}. \quad (117)$$

Furthermore, a code $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$ is said to pass the rank test by a set $\mathcal{H}$ if it passes the test for all vectors in $\mathcal{H}$.

To compare vectors, we use the concept of majorization defined in [26].

Definition 7 (Ch. 1, Def. A.1 in [26]) For a vector $\mathbf{x} = [x_1, \dots, x_n]^T$, we use $x_{(1)} \geq \dots \geq x_{(n)}$ to denote its components sorted in decreasing order. For two vectors $\mathbf{a}$ and $\mathbf{b}$ with length n , we say that $\mathbf{a}$ majorizes $\mathbf{b}$, denoted $\mathbf{a} \succeq \mathbf{b}$, if

$$\sum_{i=1}^n a_i = \sum_{i=1}^n b_i, \quad \sum_{i=1}^k a_{(i)} \geq \sum_{i=1}^k b_{(i)}, \quad \forall k \in [n]. \quad (118)$$

Remark 7 Definition 6 characterizes a structured data retrieval property of the storage code $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$. Given a vector $\mathbf{h} = [h_1, \dots, h_n]$, the test checks whether the file can be reconstructed by retrieving h_i symbols from node i , $\forall i \in [n]$, with the total number of retrieved symbols equal to B_0 . Thus, passing the test for a set $\mathcal{H}$ guarantees recoverability under column-allocation patterns specified by vectors in $\mathcal{H}$.

In our construction, the invariant set $\mathcal{H}$ consists of all integer vectors majorized by the vector $\mathbf{c}$. We prove that after every repair step, the updated storage matrices $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$ continue to pass the rank test for this entire set $\mathcal{H}$.

Theorem 3 Consider an $(n, k, d, \alpha, \beta_q, B)$ distributed storage system with

$$B = \sum_{i=0}^{k-1} \min\{2(d-i)\beta_q, d\beta_q, \alpha\}. \quad (119)$$

Define a set $\mathcal{H}$ with all vectors majorized by $\mathbf{c}$ as

$$\mathcal{H} \triangleq \{\mathbf{h} \in \mathbb{Z}_+^n | \mathbf{c} \succeq \mathbf{h}\}. \quad (120)$$

For any finite field $\mathbb{F}_q$ with

$$q > d_1 \triangleq \max \left\{ \binom{n\alpha_0}{B_0} B_0, 2|\mathcal{H}|B_0 \right\}, \quad (121)$$

there exists a coding scheme such that after every repair, the storage code for each instance, $\bar{\mathbf{Q}}_{[n]}$, passes the rank test by $\mathcal{H}$.

We present the following remarks to specify the invariant.

Remark 8 (Structure of $\mathcal{H}$) *The set $\mathcal{H}$ described in (120) consists of all positive integer vectors $\mathbf{h} = [h_1, \dots, h_n]$ with the following property*

$$\begin{aligned}
c_1 &\geq h_{(1)} \\
c_1 + c_2 &\geq h_{(1)} + h_{(2)} \\
&\vdots \\
c_1 + \dots + c_k &\geq h_{(1)} + \dots + h_{(k)} \\
&\vdots \\
c_1 + \dots + c_k &\geq h_{(1)} + \dots + h_{(n-1)} \\
c_1 + \dots + c_k &= h_{(1)} + \dots + h_{(n)}.
\end{aligned} \tag{122}$$

As an example, consider the setting in Fig. 1, where $n = 4$, $k = 2$, $d = 3$, and $\mathbf{c} = [3, 3, 0, 0]$. The corresponding set $\mathcal{H}$ consists of all permutations of the following vectors

$$[3, 3, 0, 0], \quad [3, 2, 1, 0], \quad [3, 1, 1, 1], \quad [2, 2, 2, 0], \quad [2, 2, 1, 1]. \tag{123}$$

Remark 9 (Invariant implies data retrieval) *Passing the test by $\mathcal{H}$ guarantees the data retrieval property. Indeed, since $\mathbf{c} = [c_1, \dots, c_k, 0, \dots, 0] \in \mathcal{H}$, all of its permutations also belong to $\mathcal{H}$. Therefore, for any choice of k storage nodes, there exists a corresponding selection pattern $c_{[k]}$ that specifies how many columns are chosen from each of the selected nodes. Since the rank test is satisfied for every such permutation, the resulting collection of columns has rank B_0 for each instance. Consequently, each instance allows recovery of its B_0 information symbols. Since the system consists of two independent instances of the file, the entire file of size $B = 2B_0$ is recoverable from any k nodes. Hence, maintaining the invariant over $\mathcal{H}$ implies the data retrieval property.*

The achievability proof proceeds by maintaining the above invariant throughout the repair rounds.

7.2.2 Induction Process

Initially, the coding matrix for each instance, $[\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n] \in \mathbb{F}_q^{B_0 \times n\alpha_0}$, is the generator matrix of an $(n\alpha_0, B_0)$ MDS code. By Schwartz-Zippel lemma, such code exists when the field size $q > \binom{n\alpha_0}{B_0} B_0$. Hence, by the MDS property, any selection of B_0 columns from $[\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n]$ yields a full-rank matrix. Hence, each instance of the initial code passes the rank test for each $\mathbf{h} \in \mathcal{H}$.

Now, assume that after the m th repair, the storage code $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$ passes the test for all vectors in $\mathcal{H}$. We show that there exists a repair scheme such that the updated storage after the $(m+1)$ th repair continues to pass the test for $\mathcal{H}$. Without loss of generality, suppose that node 1 fails and is replaced by a newcomer. The newcomer receives $2\beta_q$ qudits from

each of the d helper nodes $2, \dots, d+1$, and computes $\alpha = 2\alpha_0$ dits as its stored symbols. Following the same repair structure as in (84)-(93), the storage matrix at the newcomer has the form

$$\mathbf{Q}'_1 = \begin{bmatrix} \bar{\mathbf{Q}}'_1 & \mathbf{0}_{B_0 \times \alpha_0} \\ \mathbf{0}_{B_0 \times \alpha_0} & \bar{\mathbf{Q}}'_1 \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{V}}_{\text{dec}} & \mathbf{0}_{B_0 \times \alpha_0} \\ \mathbf{0}_{B_0 \times \alpha_0} & \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{V}}_{\text{dec}} \end{bmatrix}, \quad (124)$$

where $\bar{\mathbf{Q}} = [\bar{\mathbf{Q}}_2\bar{\mathbf{V}}_2, \dots, \bar{\mathbf{Q}}_{d+1}\bar{\mathbf{V}}_{d+1}] \in \mathbb{F}_q^{B_0 \times 2d\beta_q}$, and $\mathbf{V}_j \in \mathbb{F}_q^{\alpha_0 \times 2\beta_q}$, $j \in [2 : d+1]$, is the precoding matrix (to be designed) at node j .

Next, we show the existence of the parameters $\bar{\mathbf{V}}_{[2:d+1]}$ and $\bar{\mathbf{V}}_{\text{dec}}$ such that $\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_2, \dots, \bar{\mathbf{Q}}_n$ after the $(m+1)$ th repair still passes the test by $\mathcal{H}$. It suffices to guarantee

$$g \triangleq \prod_{\mathbf{h} \in \mathcal{H}} \det([\bar{\mathbf{Q}}'_1\mathbf{E}_{h_1}, \dots, \bar{\mathbf{Q}}_n\mathbf{E}_{h_n}]) \neq 0. \quad (125)$$

Since each determinant is a polynomial in the entries of $\bar{\mathbf{V}}_{[2:d+1]}, \bar{\mathbf{V}}_{\text{dec}}$ with degree at most $2B_0$, g is a polynomial of the variables with degree at most $2|\mathcal{H}|B_0$. By the Schwartz-Zippel lemma, it suffices to prove that every factor in g is itself a nonzero polynomial. Thus, for arbitrary fixed $\mathbf{h} \in \mathcal{H}$, we show that there exists a choice of $\bar{\mathbf{V}}_{[2:d+1]}$ and $\bar{\mathbf{V}}_{\text{dec}}$ such that

$$\det([\bar{\mathbf{Q}}'_1\mathbf{E}_{h_1}, \dots, \bar{\mathbf{Q}}_n\mathbf{E}_{h_n}]) \neq 0. \quad (126)$$

Now, we need the following lemma to establish a suitable redistribution of weights from the failed node to the helpers. Specifically, for any $\mathbf{h} \in \mathcal{H}$ after the $(m+1)$ th repair, there exists a $\mathbf{h}'$ after the m th repair with certain properties. The following lemma states the required properties for $\mathbf{h}'$.

Lemma 4 *For any $\mathbf{h} = [h_1, \dots, h_n] \in \mathcal{H}$, there exists*

$$\mathbf{h}' = [0, h'_2, \dots, h'_{d+1}, h_{d+2}, \dots, h_n] \in \mathcal{H}, \quad (127)$$

such that:

- P1: $h'_i \geq h_i, \forall i \in [2 : d+1]$, and the total increase in $h_2, \dots, h_{d+1}$ is h_1 , i.e.,

$$h_1 = \sum_{i=2}^{d+1} (h'_i - h_i). \quad (128)$$

- P2: The increase of each h_i does not exceed $2\beta_q$, i.e.,

$$h'_i - h_i \leq 2\beta_q, \quad \forall i \in [2 : d+1]. \quad (129)$$

The proof follows the main idea of the classical argument in [24]. For completeness, a

detailed proof is provided in Appendix B.

Remark 10 *Lemma 4 shows that for arbitrary $\mathbf{h} \in \mathcal{H}$, we can find another vector $\mathbf{h}' \in \mathcal{H}$ such that P1 and P2 are satisfied. This means that for a code $\bar{\mathbf{Q}}_1, \dots, \bar{\mathbf{Q}}_n$ passing the test by $\mathbf{h}'$ after the m th repair, we can select h_1 columns from $\bar{\mathbf{Q}}_2, \dots, \bar{\mathbf{Q}}_{d+1}$ to generate the storage of the newcomer during the $(m+1)$ th repair, $\bar{\mathbf{Q}}'_1$, such that the updated code $\bar{\mathbf{Q}}'_1, \bar{\mathbf{Q}}_2, \dots, \bar{\mathbf{Q}}_n$ can pass the test by $\mathbf{h}$.*

Since the storage after the m th repair passes the test for all vectors in $\mathcal{H}$, in particular, it passes the test for $\mathbf{h}'$. Thus,

$$\text{rank}([\bar{\mathbf{Q}}'_1 \mathbf{E}_{h'_1}, \dots, \bar{\mathbf{Q}}_n \mathbf{E}_{h_n}]) = B_0. \quad (130)$$

We now design the repair so that the newcomer reproduces exactly the h_1 columns that distinguish $\mathbf{h}$ from $\mathbf{h}'$.

Specifically, recalling (124), for each instance, the storage matrix is

$$\bar{\mathbf{Q}}'_1 = \bar{\mathbf{Q}} \mathbf{A} \bar{\mathbf{V}}_{\text{dec}} = [\bar{\mathbf{Q}}_2 \bar{\mathbf{V}}_2, \dots, \bar{\mathbf{Q}}_{d+1} \bar{\mathbf{V}}_{d+1}] \mathbf{A} \bar{\mathbf{V}}_{\text{dec}}. \quad (131)$$

Now, we design each precoding matrix $\bar{\mathbf{V}}_i, i \in [2 : d+1]$, as

$$\bar{\mathbf{V}}_i = [\mathbf{e}_1, \dots, \mathbf{e}_{h'_i - h_i}, \mathbf{0}_{\alpha_0 \times (2\beta_q - (h'_i - h_i))}], \quad (132)$$

i.e., the precoding matrix selects exactly the first $h'_i - h_i$ columns of $\bar{\mathbf{Q}}_i$. By property P1, exactly h_1 columns are selected across all helpers. Let $\mathcal{S}$ index these h_1 selected columns. As in the QMSR construction, let $\tilde{\mathbf{A}}$ be the submatrix of $\mathbf{A}$, formed by the rows indexed by $\mathcal{S}$, and set

$$\bar{\mathbf{V}}_{\text{dec}} = \tilde{\mathbf{A}}. \quad (133)$$

Then, the storage matrix for each instance at the repaired node is

$$\bar{\mathbf{Q}}'_1 = \bar{\mathbf{Q}}_{[:,\mathcal{S}]} \tilde{\mathbf{A}} \tilde{\mathbf{A}}^T. \quad (134)$$

Since $\tilde{\mathbf{A}} \tilde{\mathbf{A}}^T$ is invertible, the newcomer stores $\text{colspan}(\bar{\mathbf{Q}}_{[:,\mathcal{S}]})$, i.e., precisely the h_1 redistributed columns. Therefore, $[\bar{\mathbf{Q}}'_1 \mathbf{E}_{h_1}, \dots, \bar{\mathbf{Q}}_n \mathbf{E}_{h_n}]$ has full-rank B_0 , because it coincides with the full-rank configuration associated with $\mathbf{h}'$. Thus,

$$\det([\bar{\mathbf{Q}}'_1 \mathbf{E}_{h_1}, \dots, \bar{\mathbf{Q}}_n \mathbf{E}_{h_n}]) \neq 0, \quad (135)$$

which the determinant for this fixed $\mathbf{h}$ is a nonzero polynomial.

Therefore, each determinant in the product in (125) is a nonzero polynomial. The total

degree is at most d_1 . Schwartz-Zippel lemma guarantees the existence of a common assignment of $\bar{\mathbf{V}}_{[2:d+1]}$ and $\bar{\mathbf{V}}_{\text{dec}}$ such that the storage after the $(m+1)$ th repair passes the test for *all* $\mathbf{h} \in \mathcal{H}$. Therefore, the invariant is preserved under every repair, completing the induction and establishing achievability for the general point on the tradeoff curve.

8 Conclusion

In this paper, we investigated the fundamental limits of distributed storage systems with quantum-assisted repair. We fully characterized the storage-bandwidth tradeoff enabled by quantum communication and established its optimality through matching converse and achievability proofs. The converse argument relies on quantum information theoretic inequalities, while the achievability proof develops a unified framework based on maintaining a structured rank invariant.

Our results show that quantum entanglement can strictly improve the classical storage-bandwidth tradeoff in general. In particular, when $d \geq 2k - 2$, we identified an optimal operating point that lies outside the classical tradeoff region and simultaneously minimizes the storage and repair bandwidth. This improvement arises from the ability of entanglement-assisted repair to transmit classical information more efficiently than purely classical schemes.

To the best of our knowledge, this work initiates the study of quantum-assisted distributed storage systems. This leads to many interesting open problems. For instance, if the newcomer is allowed to share prior entanglement with the helper nodes, the transmitted qudits can potentially carry more classical information via superdense coding, which may further enlarge the achievable tradeoff region. Thus, characterizing the resulting storage-bandwidth tradeoff remains an open problem. In addition, this work focuses on functional regeneration in the quantum setting. It would be of significant interest to determine the optimal tradeoff for exact regeneration under quantum-assisted repair, since the complete characterization of the exact-regeneration tradeoff remains unresolved even in the classical setting [27, 28].

Appendix

A Proof of Remark 4

Proof: Throughout the proof, we restrict to integers d satisfying $k \leq d \leq 2k - 2$. Let $t \triangleq \lfloor d/2 \rfloor$. Note that the denominator of $r(d, k)$ equals

$$2d(t+1) + 2(2d - k - t)(k - t - 1), \quad (136)$$

where the two terms are both nonnegative in this regime.

When $d = 2m$ or $d = 2m + 1$, we have $t = m$. Then,

$$r(2m, k) = \frac{k(4m - k + 1)}{D_{2m}}, \quad (137)$$

$$r(2m + 1, k) = \frac{k(4m - k + 3)}{D_{2m+1}}, \quad (138)$$

where

$$D_{2m} \triangleq -2k^2 + 8km + 2k - 2m^2 - 2m, \quad (139)$$

$$D_{2m+1} \triangleq -2k^2 + 8km + 6k - 2m^2 - 4m - 2. \quad (140)$$

Similarly, replacing m by $m + 1$ in (137)-(139) gives

$$r(2m + 2, k) = \frac{k(4m - k + 5)}{D_{2m+2}}, \quad (141)$$

where

$$D_{2m+2} \triangleq -2k^2 + 8k(m + 1) + 2k - 2(m + 1)^2 - 2(m + 1). \quad (142)$$

We now show that

$$r(2m + 1, k) \geq r(2m, k), \quad r(2m + 2, k) \geq r(2m + 1, k) \quad (143)$$

whenever $k \leq 2m$ and $2m + 2 \leq 2k - 2$. Using (137)-(140), we have

$$r(2m + 1, k) - r(2m, k) = \frac{2k(m + 1)(2m + 1 - k)}{D_{2m} D_{2m+1}}. \quad (144)$$

Since $k \leq 2m$ implies $2m + 1 - k \geq 1$, and all denominators are positive, the right-hand side of (144) is nonnegative, proving $r(2m + 1, k) \geq r(2m, k)$. Likewise,

$$r(2m + 2, k) - r(2m + 1, k) = \frac{2k(m + 1)(2m + 1 - k)}{D_{2m+1} D_{2m+2}} \geq 0, \quad (145)$$

which proves the second inequality in (143). Furthermore, by (143), $r(d, k)$ is monotonically increasing as d increases by 1, hence, it is monotonically increasing over the entire integer interval $k \leq d \leq 2k - 2$. We can evaluate $r(2k - 2, k)$ as the maximal value. Since $2k - 2$ is even, we write $2k - 2 = 2m$ with $m = k - 1$ and apply (137),

$$r(2k - 2, k) = \frac{k(4(k - 1) - k + 1)}{D_{2(k-1)}} = \frac{3k(k - 1)}{4k(k - 1)} = \frac{3}{4}. \quad (146)$$

This proves the property 1.

For property 2, let $\Delta_1 \triangleq r(2m+1, k) - r(2m, k)$ and $\Delta_2 \triangleq r(2m+2, k) - r(2m+1, k)$. From (144)-(145), we have

$$\Delta_1 = \frac{2k(m+1)(2m+1-k)}{D_{2m}D_{2m+1}}, \quad (147)$$

$$\Delta_2 = \frac{2k(m+1)(2m+1-k)}{D_{2m+1}D_{2m+2}}. \quad (148)$$

To show $\Delta_1 \geq \Delta_2$, it is sufficient to show that

$$\frac{1}{D_{2m}} \geq \frac{1}{D_{2m+2}}, \quad \text{i.e., } D_{2m} \leq D_{2m+2}. \quad (149)$$

Using (139) and (142), we compute

$$\begin{aligned} D_{2m+2} - D_{2m} &= (-2k^2 + 8k(m+1) + 2k - 2(m+1)^2 - 2(m+1)) \\ &\quad - (-2k^2 + 8km + 2k - 2m^2 - 2m) \end{aligned} \quad (150)$$

$$= 8k - 4m - 4 = 4(2k - m - 1). \quad (151)$$

Since $2m+2 \leq 2k-2$ implies $m \leq k-2$, we have $2k - m - 1 \geq k + 1 > 0$, hence $D_{2m+2} - D_{2m} > 0$. Together with (149), this yields $\Delta_1 \geq \Delta_2$, i.e.,

$$r(2m+1, k) - r(2m, k) \geq r(2m+2, k) - r(2m+1, k), \quad (152)$$

which proves the property 2. This completes the proof. ■

B Proof of Lemma 4

The idea of the proof is to trace back the quantity h_1 from the newcomer (treated as node 1) to the helper nodes $2, \dots, d+1$ by a brick-moving process. At each step, one unit (brick) is removed from h_1 and assigned to one of the d helper nodes with the current minimum value. For example, when $\mathbf{h} = [3, 3, 0, 0]$, $d = 2$, and the newcomer communicates with nodes 3 and 4, the redistribution can be carried out as

$$\mathbf{h} = [3, 3, 0, 0] \rightarrow [2, 3, 1, 0] \rightarrow [1, 3, 1, 1] \rightarrow [0, 3, 2, 1] = \mathbf{h}'. \quad (153)$$

Since $h_1 \leq c_1 \leq d\beta_q$, at most $d\beta_q$ bricks are removed from coordinate 1. These bricks are distributed among the d coordinates $2, \dots, d+1$. As a result, the two properties P1 and P2 can be satisfied by each of $h_2, \dots, h_{d+1}$ receiving at most $2\beta_q$ symbols when we are totally assigning h_1 symbols to them.

Now, it remains to show $\mathbf{h}' \in \mathcal{H}$. Specifically, we conduct the brick-moving process as

$$\mathbf{h} = \mathbf{h}^{(0)} \rightarrow \mathbf{h}^{(1)} \rightarrow \dots \rightarrow \mathbf{h}^{(h_1)} = \mathbf{h}', \quad (154)$$

which is the sequence of vectors generated by the brick-moving process, where each step transfers one brick from coordinate 1 to a coordinate in $[2 : d + 1]$. We will show that

$$\mathbf{h}^{(i)} \in \mathcal{H}, \quad \forall i = [0 : h_1]. \quad (155)$$

Suppose, for contradiction, that there exists a smallest index $i \geq 1$ such that $\mathbf{h}^{(i-1)} \in \mathcal{H}$ but $\mathbf{h}^{(i)} \notin \mathcal{H}$. It is important to note that from the definition, we have $c_1 \geq c_2 \geq \dots \geq c_k$. By the definition of $\mathcal{H}$ and (122), this implies that there exists an ℓ with $\ell < k$ such that

$$h_{(1)}^{(i)} + \dots + h_{(\ell)}^{(i)} > c_1 + \dots + c_\ell. \quad (156)$$

Without loss of generality, assume that ℓ is the smallest index for which this inequality holds. Let $i_1, \dots, i_\ell$ be the indices of the ℓ largest entries of $\mathbf{h}^{(i)}$. We distinguish two cases according to whether $h_1^{(i)}$ is in this index set, i.e., whether $1 \in \{i_1, \dots, i_\ell\}$. We show that both cases lead to a contradiction.

B.1 Case 1: $1 \in \{i_1, \dots, i_\ell\}$

In this case, the $h_1^{(i)}$ is in the top- ℓ entries in $\mathbf{h}^{(i)}$. Recall that in each step of the brick-moving process, the value at coordinate 1 decreases by exactly one, while all other coordinates are nondecreasing. Hence, for any fixed index set that includes 1, the sum of the corresponding entries is nonincreasing over steps, i.e., $h_{i_1}^{(j)} + \dots + h_{i_\ell}^{(j)}$ is nonincreasing with $j = 0, \dots, h_1$. Therefore,

$$h_{i_1}^{(0)} + \dots + h_{i_\ell}^{(0)} \geq h_{i_1}^{(i)} + \dots + h_{i_\ell}^{(i)}. \quad (157)$$

By the choice of ℓ as the smallest index for which (156) is violated, we further have

$$h_{i_1}^{(i)} + \dots + h_{i_\ell}^{(i)} > c_1 + \dots + c_\ell. \quad (158)$$

Combining the above two inequalities yields

$$h_{i_1} + \dots + h_{i_\ell} = h_{i_1}^{(0)} + \dots + h_{i_\ell}^{(0)} > c_1 + \dots + c_\ell. \quad (159)$$

Hence, the sum of the top ℓ elements of $\mathbf{h}^{(0)}$ is greater than the $c_1 + \dots + c_\ell$. This contradicts the assumption that the initial vector $\mathbf{h} = \mathbf{h}^{(0)} \in \mathcal{H}$.

B.2 Case 2: $1 \notin \{i_1, \dots, i_\ell\}$

In this case, $h_1^{(i)}$ is not among the ℓ largest entries in $\mathbf{h}^{(i)}$. Now, let ℓ' be the *largest* index such that

$$h_{(\ell')}^{(i)} \geq h_{(\ell)}^{(i)} - 1, \quad (160)$$

which means that the drop from $h_{(\ell)}^{(i)}$ to $h_{(\ell')}^{(i)}$ is at most one, and ℓ' is chosen maximal with this property. Since such ℓ' is not guaranteed to be larger than ℓ , we consider two subcases: $\ell' = \ell$ and $\ell' > \ell$. We will show that both lead to a contraction.

Consider the case of $\ell' = \ell$, i.e., the gap between $h_{(\ell)}^{(i)}$ and $h_{(\ell+1)}^{(i)}$ is at least 2. Recall that $\mathbf{h}^{(i-1)} \in \mathcal{H}$ and $\mathbf{h}^{(i)} \notin \mathcal{H}$. The violation at ℓ must be caused by an increase of one of $h_{i_1}^{(i-1)}, \dots, h_{i_\ell}^{(i-1)}$. This implies that at least $d - \ell$ slots have been filled out of the index set $\{i_1, \dots, i_\ell\}$, i.e., at least $2(d - \ell)\beta_{\mathbf{q}}$ bricks have been moved from h_1 . Hence, we have

$$h_{i_1} + \dots + h_{i_\ell} + h_1 \geq h_{i_1}^{(i)} + \dots + h_{i_\ell}^{(i)} + 2(d - \ell)\beta_{\mathbf{q}} \quad (161)$$

$$> c_1 + \dots + c_\ell + 2(d - \ell)\beta_{\mathbf{q}} \quad (162)$$

$$\geq c_1 + \dots + c_\ell + c_{\ell+1}, \quad (163)$$

where the last inequality is because $c_{\ell+1} = \min\{d\beta_{\mathbf{q}}, 2(d - \ell)\beta_{\mathbf{q}}, \alpha\} \leq 2(d - \ell)\beta_{\mathbf{q}}$. The inequality (163) contradicts $\mathbf{h} \in \mathcal{H}$ due to (162).

In the case of $\ell' > \ell$, we first claim that

$$h_{(1)}^{(i)} + \dots + h_{(\ell')}^{(i)} > c_1 + \dots + c_{\ell'}, \quad (164)$$

i.e., ℓ' also violates the majorization property.

To prove the claim, note that from the property of ℓ , we have

$$h_{(1)}^{(i)} + \dots + h_{(\ell-1)}^{(i)} \leq c_1 + \dots + c_{\ell-1}, \quad (165)$$

$$h_{(1)}^{(i)} + \dots + h_{(\ell-1)}^{(i)} + h_{(\ell)}^{(i)} > c_1 + \dots + c_{\ell-1} + c_\ell, \quad (166)$$

which means $h_{(\ell)}^{(i)} > c_\ell$. Meanwhile, for ℓ' , we have

$$h_{(\ell+1)}^{(i)} \geq \dots \geq h_{(\ell')}^{(i)} \geq h_{(\ell)}^{(i)} - 1 \geq c_\ell. \quad (167)$$

Thus,

$$h_{(1)}^{(i)} + \dots + h_{(\ell)}^{(i)} + (h_{(\ell+1)}^{(i)} + \dots + h_{(\ell')}^{(i)}) > c_1 + \dots + c_\ell + (h_{(\ell+1)}^{(i)} + \dots + h_{(\ell')}^{(i)}) \quad (168)$$

$$\geq c_1 + \dots + c_\ell + (\ell' - \ell)c_\ell \quad (169)$$

$$\geq c_1 + \dots + c_\ell + (c_{\ell+1} + \dots + c_{\ell'}). \quad (170)$$

This proves the claim.

Since $1 \notin i_{[\ell]}$ in this section, we consider the cases of $1 \in i_{[\ell:\ell']}$ and $1 \in i_{[\ell'+1:n]}$, respectively. When $1 \in i_{[\ell:\ell']}$, since ℓ' also violates the majorization property, we have

$$h_{i_1} + \dots + h_{i_{\ell'}} \geq h_{i_1}^{(i)} + \dots + h_{i_{\ell'}}^{(i)} > c_1 + \dots + c_{\ell'}. \quad (171)$$

On the other hand, when $1 \in i_{[\ell'+1:n]}$, we have

$$h_{i_1} + \dots + h_{i_{\ell'}} + h_1 \geq h_{i_1}^{(i)} + \dots + h_{i_{\ell'}}^{(i)} + 2(d - \ell')\beta_q \quad (172)$$

$$> c_1 + \dots + c_{\ell'} + 2(d - \ell')\beta_q \quad (173)$$

$$\geq c_1 + \dots + c_{\ell'} + c_{\ell'+1}, \quad (174)$$

where (172) follows the same reason as the inequality (161) that at least $2(d - \ell')\beta_q$ bricks have been moved from h_1 , and the inequality (174) is because $c_{\ell'+1} \triangleq \min\{d\beta_q, 2(d - \ell')\beta_q, \alpha\} \leq 2(d - \ell')\beta_q$.

Therefore, all these cases contradict $\mathbf{h} \in \mathcal{H}$. This completes the proof of Lemma 4.

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